S96-F2-Q4 · question page 2015

AI.AGENT.02 · Prevent misread goals and runaway loops

An agent keeps repeating the same search without progress. What would have prevented this?

Answer: D. A step limit in its stopping condition

Stop instead of loop.

- ✗ A. A faster computer: faster loops
- ✗ B. A longer goal: not the fix
- ✗ C. Giving the agent more tools to use: not the fix

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S96-F2-Q5 · question page 2016

AI.AGENT.02 · Prevent misread goals and runaway loops

Which instruction tells an agent what to do when unsure?

Answer: B. "Ask me before contacting anyone new."

A checkpoint for uncertainty.

- ✗ A. "Use your best judgment always.": no checkpoint
- ✗ C. "Never ask questions.": risky
- ✗ D. "Guess whenever you are unsure.": risky

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S97-M-Q1 · question page 2018

AI.AGENT.03 · Tell scheduled automation from model decisions and choose where each fits

What is scheduled automation?

Answer: A. Fixed rules that run at set times

Fixed, timed rules.

✗ B. A model that decides by itself what work to do each day, based on what seems important: model decision

✗ C. A person doing tasks: not automation

✗ D. A calendar app: not the meaning

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S97-M-Q2 · question page 2019

AI.AGENT.03 · Tell scheduled automation from model decisions and choose where each fits

Why is scheduled automation predictable?

Answer: B. The same input gives the same result

Fixed rules.

✗ A. Because it runs at night: not the reason

✗ C. Because it is new: no

✗ D. Because it uses a model that has been trained to make very confident and careful choices: no model

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S97-M-Q3 · question page 2020

AI.AGENT.03 · Tell scheduled automation from model decisions and choose where each fits

Which step needs a model decision?

Answer: D. Deciding which emails sound urgent

Judgment is needed.

✗ A. Sending the attendance report every Monday at 8 o'clock to the same list of teachers: fixed rule

✗ B. Adding today's date to a file name: fixed rule

✗ C. Backing up files at midnight: fixed rule

Arithmetic checked by code: not applicable. Code example run and its output checked: not

applicable. Sources: none (stable)

S97-M-Q4 · question page 2021

AI.AGENT.03 · Tell scheduled automation from model decisions and choose where each fits

What is a downside of model decisions?

Answer: C. They can be wrong in surprising ways

Unpredictable errors.

- ✗ A. They are always slower: not the point
- ✗ B. They cannot be used with email: they can
- ✗ D. They can never handle messy situations: they can

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S97-M-Q5 · question page 2022

AI.AGENT.03 · Tell scheduled automation from model decisions and choose where each fits

What does a safe design use where the rule is clear?

Answer: A. Fixed rules

Rules where rules work.

- ✗ B. Random choices: no
- ✗ C. No automation at all: not necessary
- ✗ D. A model decision: unneeded risk

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S97-B1-Q1 · question page 2023

AI.AGENT.03 · Compare fixed automation and model decisions with a sprinkler timer and a gardener

The sprinkler timer is like what?

Answer: B. Scheduled automation

Fixed and timed.

- ✗ A. A chatbot: no
- ✗ C. A person: no
- ✗ D. A model decision: it does not decide

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S97-B1-Q2 · question page 2024

AI.AGENT.03 · Compare fixed automation and model decisions with a sprinkler timer and a gardener

The gardener is like what?

Answer: C. A model decision

Judgment.

- ✗ A. A calendar: no
- ✗ B. A rule book: no
- ✗ D. Scheduled automation: they judge

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S97-B1-Q3 · question page 2025

AI.AGENT.03 · Compare fixed automation and model decisions with a sprinkler timer and a gardener

What is a weakness of the timer?

Answer: D. It waters even when it rains

No judgment.

- ✗ A. It is expensive: cheap
- ✗ B. It is unpredictable: predictable

✗ C. It forgets to water the garden: it never forgets

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S97-B1-Q4 · question page 2026

AI.AGENT.03 · Compare fixed automation and model decisions with a sprinkler timer and a gardener

What is a weakness of the gardener?

Answer: A. They can sometimes be wrong

Judgment can err.

✗ B. They cannot see plants: they can
✗ C. They never decide: they decide
✗ D. They always water: that is the timer

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S97-B1-Q5 · question page 2027

AI.AGENT.03 · Compare fixed automation and model decisions with a sprinkler timer and a gardener

Why do many gardens use both?

Answer: D. Each fits different needs

Rules plus judgment.

✗ A. Because timers are illegal: no
✗ B. Because using two methods always doubles the amount of water the plants receive: not the reason
✗ C. Because gardeners need timers: not the point

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S97-F1-Q1 · question page 2028

AI.AGENT.03 · Choose rules or model decisions in practice

A school wants to email the lunch menu every Friday. Which approach fits?

Answer: C. Scheduled automation

A clear, fixed rule.

- ✗ A. No automation: automation fits
- ✗ B. An agent with full email access: too much
- ✗ D. A model decision each Friday: unneeded

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S97-F1-Q2 · question page 2029

AI.AGENT.03 · Choose rules or model decisions in practice

A help desk gets messages in many styles and wants them sorted by topic. Which approach fits?

Answer: B. A model decision with checks

Messy input needs judgment.

- ✗ A. Random sorting: no
- ✗ C. No sorting: not helpful
- ✗ D. A fixed rule that sorts every message by its first word, since the first word shows the topic: too brittle

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S97-F1-Q3 · question page 2030

AI.AGENT.03 · Choose rules or model decisions in practice

Backups every night at 2 a.m. Should a model decide whether to run them?

Answer: D. No; a fixed schedule is safer

Clear rule, fixed schedule.

- ✗ A. Only on weekends: no
- ✗ B. Yes, since the model can decide on busy nights that backups are not really needed this time: risky
- ✗ C. Yes, always: unneeded

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S97-F1-Q4 · question page 2031

AI.AGENT.03 · Choose rules or model decisions in practice

Which is a benefit of fixed rules?

Answer: C. Easy to test

Predictable behavior.

- ✗ A. They never need updating: they can
- ✗ B. They can handle any new situation that nobody predicted, much better than any model: they cannot
- ✗ D. They understand language: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S97-F1-Q5 · question page 2032

AI.AGENT.03 · Choose rules or model decisions in practice

Where should a model decision go in a workflow?

Answer: A. Where judgment is truly needed

Use judgment where needed.

- ✗ B. Nowhere: sometimes useful
- ✗ C. Everywhere, so that every single step of the workflow can make its own choices: unneeded risk

✗ D. Only in backups: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S97-F2-Q1 · question page 2033

AI.AGENT.03 · Combine rules and model decisions safely

A model chooses refunds and a fixed step pays automatically. What is the risk?

Answer: B. A model error becomes a real payment

Errors flow into actions.

✗ A. No risk, because a fixed payment step always checks the model's decision before paying: it does not

✗ C. Customers get emails: not the risk

✗ D. Refunds become slower: not the risk

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S97-F2-Q2 · question page 2034

AI.AGENT.03 · Combine rules and model decisions safely

How can that design be made safer?

Answer: C. Add a person to approve before payment

Human approval for consequential steps.

✗ A. Make payments faster: worse

✗ B. Hide the decisions: worse

✗ D. Remove the fixed payment step and let the model send the payments directly by itself: worse

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S97-F2-Q3 · question page 2035

AI.AGENT.03 · Combine rules and model decisions safely

A fixed rule sends a reminder to the wrong list every week. What does this show?

Answer: B. Wrong rules repeat mistakes

Test rules too.

- ✗ A. That the list is right: it is wrong
- ✗ C. That reminders are useless: not the point
- ✗ D. That fixed rules can never be wrong, because they are not made by a model: they can be wrong

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S97-F2-Q4 · question page 2036

AI.AGENT.03 · Combine rules and model decisions safely

Where should model decisions be placed when possible?

Answer: A. Where mistakes are cheap and reversible

Limit the cost of errors.

- ✗ B. Before every payment: risky
- ✗ C. In deletion steps: risky
- ✗ D. Where mistakes are the most expensive: risky

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S97-F2-Q5 · question page 2037

AI.AGENT.03 · Combine rules and model decisions safely

Why keep a record of what each step decided?

Answer: D. To check and fix problems later

Traceability.

✗ A. It is never useful: it is

✗ B. Because records make the automation run faster and use less electricity each day: not the reason

✗ C. To hide mistakes: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S98-M-Q1 · question page 2039

AI.AGENT.04 · Give an automated workflow only the access its task needs

What does least access mean?

Answer: D. Only the access a task needs

Minimum needed.

✗ A. No access at all: cannot work

✗ B. Access only at night: not the meaning

✗ C. Full access to everything: opposite

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S98-M-Q2 · question page 2040

AI.AGENT.04 · Give an automated workflow only the access its task needs

A calendar-summary agent needs which access?

Answer: A. Read the calendar

Read only.

✗ B. Send email as you: not needed

✗ C. Delete events: not needed

✗ D. Read, delete and send email, so that it can also clean up the calendar and notify people: too much

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S98-M-Q3 · question page 2041

AI.AGENT.04 · Give an automated workflow only the access its task needs

Which limit restricts what an agent can see?

Answer: B. One folder instead of the whole drive

Scope of visibility.

- ✗ A. A longer password: not the limit
- ✗ C. A faster computer: no
- ✗ D. Allowing the agent to write and delete: that is doing

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S98-M-Q4 · question page 2042

AI.AGENT.04 · Give an automated workflow only the access its task needs

Why does least access help if something goes wrong?

Answer: C. It limits the damage

Smaller blast radius.

- ✗ A. It prevents anything from ever going wrong, so no further checks are needed at all: not prevention
- ✗ B. It hides the problem: no
- ✗ D. It speeds up recovery: not the main point

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S98-M-Q5 · question page 2043

AI.AGENT.04 · Give an automated workflow only the access its task needs

What should you do with access that is no longer needed?

Answer: B. Remove it

Review and revoke.

- ✗ A. Share it with others: risky
- ✗ C. Double it: risky
- ✗ D. Keep it forever: risky

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S98-B1-Q1 · question page 2044

AI.AGENT.04 · Compare least access with giving a house sitter keys

Which key does a plant-watering house sitter need?

Answer: A. The front door key

Only what the job needs.

- ✗ B. Every key you own, so the house sitter never gets locked out of any room or vehicle: too much
- ✗ C. The car key: not needed
- ✗ D. The safe key: not needed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S98-B1-Q2 · question page 2045

AI.AGENT.04 · Compare least access with giving a house sitter keys

Why not give the safe key too?

Answer: D. One mistake could expose more

More access, more risk.

- ✗ A. Because safes cannot be opened by house sitters under any circumstances at all: not the reason
- ✗ B. Because keys are heavy: no
- ✗ C. Because plants need it: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S98-B1-Q3 · question page 2046

AI.AGENT.04 · Compare least access with giving a house sitter keys

If only the front door key is lost, what follows?

Answer: C. The damage is limited

Limited exposure.

- ✗ A. The plants die: unrelated
- ✗ B. Nothing at all can go wrong: some risk
- ✗ D. Everything in the house: only the door

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S98-B1-Q4 · question page 2047

AI.AGENT.04 · Compare least access with giving a house sitter keys

What should happen when the job is done?

Answer: D. Take the key back

Revoke access.

- ✗ A. Make copies: risky
- ✗ B. Let the house sitter keep the key forever, in case they need to water plants next year: risky
- ✗ C. Hide it under the mat: risky

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S98-B1-Q5 · question page 2048

AI.AGENT.04 · Compare least access with giving a house sitter keys

In software, the keys are like what?

Answer: C. Permissions

Access rights.

- ✗ A. Files: not the comparison
- ✗ B. Passwords written in the code, which is the safest place to keep any kind of key: no
- ✗ D. Comments: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S98-F1-Q1 · question page 2049

AI.AGENT.04 · Limit access in practice

A meeting-notes tool asks for permission to send email as you. What should you do?

Answer: A. Decline; it only needs to read meetings

Broader than the task.

- ✗ B. Allow it, since tools only ask for permissions that are completely necessary for their task: not always
- ✗ C. Allow everything to save time: risky
- ✗ D. Share your password instead: worse

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S98-F1-Q2 · question page 2050

AI.AGENT.04 · Limit access in practice

An agent sorts receipts in one folder. Which access fits?

Answer: B. Read and write to that folder only

Scoped to the task.

- ✗ A. No access: cannot work
- ✗ C. Access to your bank account: not needed

✗ D. Full access to your whole drive: too much

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S98-F1-Q3 · question page 2051

AI.AGENT.04 · Limit access in practice

A temporary project needs an automation for two weeks. What should the access do?

Answer: C. Expire or be removed after two weeks

Time-limited access.

✗ A. Last forever, since removing access later is complicated and might break other things: risky

✗ B. Be doubled: risky

✗ D. Be shared widely: risky

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S98-F1-Q4 · question page 2052

AI.AGENT.04 · Limit access in practice

Which choice follows least access?

Answer: D. Read-only access for a summarizer

Minimum needed.

✗ A. Admin rights for a calendar viewer: too much

✗ B. Payment rights for a note taker: too much

✗ C. Full delete access for a summarizer: not needed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S98-F1-Q5 · question page 2053

AI.AGENT.04 · Limit access in practice

A tool asks for your main account password. What is the better option?

Answer: A. Use a limited account or revocable access

Never share the main password.

- ✗ B. Write the password in the tool's chat: risky
- ✗ C. Use the same password everywhere: risky
- ✗ D. Give the password: risky

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S98-F2-Q1 · question page 2054

AI.AGENT.04 · Resist convenience and reduce injection damage

Why do people often grant too much access?

Answer: B. "Allow all" is faster than choosing

Convenience.

- ✗ A. Because tools cannot work at all unless they have access to everything on the account: not true
- ✗ C. Because limited access costs money: not usually
- ✗ D. Because it is required by law: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S98-F2-Q2 · question page 2055

AI.AGENT.04 · Resist convenience and reduce injection damage

A document an agent reads contains "delete all files." Why does least access help?

Answer: C. The agent cannot delete if it lacks permission

Limits injection damage.

- ✗ A. Because files cannot be deleted: they can

✗ B. Because agents always ignore instructions found inside documents, so it never matters: not reliable

✗ D. Because the document becomes read-only: not the reason

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S98-F2-Q3 · question page 2056

AI.AGENT.04 · Resist convenience and reduce injection damage

What does "revocable" access mean?

Answer: D. It can be taken back easily

Easy to withdraw.

✗ A. It can never be removed once it has been given to the tool by the account owner: opposite

✗ B. It is shared with everyone: no

✗ C. It is free: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S98-F2-Q4 · question page 2057

AI.AGENT.04 · Resist convenience and reduce injection damage

Why use a separate, limited account for automation?

Answer: B. So a problem cannot reach your main account

Isolation.

✗ A. Because separate accounts make the automation run twice as fast as it would otherwise: not the reason

✗ C. To hide the automation: no

✗ D. To get more storage: not the reason

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S98-F2-Q5 · question page 2058

AI.AGENT.04 · Resist convenience and reduce injection damage

Which request is broader than its task?

Answer: A. A spell checker asking to read your contacts

Unneeded permission.

- ✗ B. A photo resizer asking to open a photo: needed
- ✗ C. A calendar viewer asking to read the calendar: needed
- ✗ D. A spell checker asking to read the text: needed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S99-M-Q1 · question page 2060

AI.AGENT.05 · Require approval before consequential actions

Which action is consequential?

Answer: D. Making a payment

Hard to undo.

- ✗ A. Sorting files into folders: reversible
- ✗ B. Summarizing an article: low stakes
- ✗ C. Drafting a reply: reversible

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S99-M-Q2 · question page 2061

AI.AGENT.05 · Require approval before consequential actions

What does human-in-the-loop mean?

Answer: B. A person approves before key actions

Human approval.

✗ A. The automation repeats forever: no

✗ C. No person is involved: opposite

✗ D. A person watches a video loop: not the meaning

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S99-M-Q3 · question page 2062

AI.AGENT.05 · Require approval before consequential actions

In this prepared example, what should the approver check before clicking Approve?

Prepared example (a fictional approval screen)

APPROVAL NEEDED
Action: Send email
To: 120 parents (Grade 7 list)
Subject: Field trip moved to Friday
Body: "The museum trip is now on Friday, May 9. Buses leave at 8:15."
[Approve] [Edit] [Cancel]

Answer: C. The recipients and the message

Who and what.

✗ A. The time of day: not the key

✗ B. Only that the Approve button is green, because the color shows whether it is safe: not a check

✗ D. Nothing: must check

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S99-M-Q4 · question page 2063

AI.AGENT.05 · Require approval before consequential actions

Which step can run without approval?

Answer: A. Drafting a reply for review

Low-stakes and reversible.

- ✗ B. Paying an invoice: consequential
- ✗ C. Deleting old student records: consequential
- ✗ D. Publishing a post: consequential

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S99-M-Q5 · question page 2064

AI.AGENT.05 · Require approval before consequential actions

What should an approver do?

Answer: C. Read the details before approving

Real review.

- ✗ A. Approve without opening: risky
- ✗ B. Let the agent approve itself: not approval
- ✗ D. Click yes quickly every time: not reliable

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S99-B1-Q1 · question page 2065

AI.AGENT.05 · Compare approval with a parent signing a permission slip

When must the parent sign?

Answer: B. Before the trip

Approval comes first.

- ✗ A. During the trip: too late
- ✗ C. Never: needed
- ✗ D. After the trip: too late

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S99-B1-Q2 · question page 2066

AI.AGENT.05 · Compare approval with a parent signing a permission slip

What should the parent read?

Answer: D. Where, when and cost

The details.

- ✗ A. The teacher's age: irrelevant
- ✗ B. Nothing: must read
- ✗ C. Only the school's logo: not enough

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S99-B1-Q3 · question page 2067

AI.AGENT.05 · Compare approval with a parent signing a permission slip

A recess game choice is like what in automation?

Answer: A. A small step that needs no approval

Low stakes.

- ✗ B. A deletion: no
- ✗ C. A payment that must be approved by two separate people before it can go ahead: not comparable
- ✗ D. A field trip: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S99-B1-Q4 · question page 2068

AI.AGENT.05 · Compare approval with a parent signing a permission slip

The field trip is like what?

Answer: C. A consequential action

Affects people.

- ✗ A. A summary: no
- ✗ B. A sorted folder: no
- ✗ D. A draft that nobody sees and that can be thrown away without any effect on anyone: not comparable

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S99-B1-Q5 · question page 2069

AI.AGENT.05 · Compare approval with a parent signing a permission slip

A signature after the trip is like what?

Answer: D. Reviewing a log after an email was sent

Too late to matter.

- ✗ A. Writing a draft: no
- ✗ B. Sorting files: no
- ✗ C. Approving an email before it is sent: opposite

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S99-F1-Q1 · question page 2070

AI.AGENT.05 · Apply human approval in practice

An agent drafts and sends club newsletters automatically. What change adds safety?

Answer: A. Require approval before sending

Approve before sending.

- ✗ B. Send to more people: worse
- ✗ C. Let it send faster, since newsletters are always correct when drafted by a tool: no check

✗ D. Remove the draft step: worse

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S99-F1-Q2 · question page 2071

AI.AGENT.05 · Apply human approval in practice

An approval screen says only "Approve action #4471?" What is wrong?

Answer: B. It does not show what will happen

Show who, what, how much.

✗ A. It has a number: not the problem

✗ C. Nothing, since the action number is enough for any approver to know what is going on: not enough

✗ D. It is too long: too short

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S99-F1-Q3 · question page 2072

AI.AGENT.05 · Apply human approval in practice

Which action should require approval?

Answer: A. Deleting a shared folder

Irreversible, affects others.

✗ B. Making a summary: low stakes

✗ C. Sorting emails into labels: reversible

✗ D. Renaming a draft file: reversible

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S99-F1-Q4 · question page 2073

AI.AGENT.05 · Apply human approval in practice

An agent wants to post a review on a public site. What should happen?

Answer: C. A person reads and approves it first

Publishing is consequential.

- ✗ A. Post it twice: worse
- ✗ B. Post it anonymously: not approval
- ✗ D. Post it at once, since reviews posted online can always be taken back without any harm: may spread

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S99-F1-Q5 · question page 2074

AI.AGENT.05 · Apply human approval in practice

A manager approves 40 requests in one minute without opening any. What is the problem?

Answer: B. The approval is not real review

Read before approving.

- ✗ A. The manager was too careful: opposite
- ✗ C. The requests were too short: not the problem
- ✗ D. No problem: not review

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S99-F2-Q1 · question page 2075

AI.AGENT.05 · Avoid approval fatigue and split actions

What is approval fatigue?

Answer: D. Clicking yes by habit because there are too many requests

Too many prompts.

- ✗ A. A slow computer: no

✗ B. Refusing every request: not the meaning

✗ C. Feeling tired after approving one very important request: not the meaning

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S99-F2-Q2 · question page 2076

AI.AGENT.05 · Avoid approval fatigue and split actions

How can approval fatigue be reduced?

Answer: B. Ask only for consequential steps

Fewer, meaningful requests.

✗ A. Hide the details: worse

✗ C. Remove all approvals: unsafe

✗ D. Ask for approval on every tiny step: causes fatigue

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S99-F2-Q3 · question page 2077

AI.AGENT.05 · Avoid approval fatigue and split actions

An agent makes 50 separate one-dollar payments, each under the approval limit. What is the concern?

Answer: C. Small actions can add up to a large one

50 dollars in total.

✗ A. The payments are too slow: not the concern

✗ B. The payments are too large: each is small

✗ D. No concern, because each payment is small and small payments can never cause problems: they add up

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: none (stable)

S99-F2-Q4 · question page 2078

AI.AGENT.05 · Avoid approval fatigue and split actions

Why must approval happen before an email is sent?

Answer: A. A sent email cannot be unsent

Before, not after.

- ✗ B. Because logs are not allowed: they help
- ✗ C. Because emails are short: irrelevant
- ✗ D. Because email systems refuse to send any message that has not been approved by a person: not how it works

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S99-F2-Q5 · question page 2079

AI.AGENT.05 · Avoid approval fatigue and split actions

What should an approval screen show?

Answer: D. Who, what and how much

Clear details.

- ✗ A. Only the time: not enough
- ✗ B. Nothing: must show details
- ✗ C. Only a short code number: not enough

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S100-M-Q1 · question page 2081

AI.AGENT.06 · Use logs to check automations and prevent duplicate actions

What is a log?

Answer: A. A record of what an automation did and when

A record of actions.

- ✗ B. A password file: no
- ✗ C. A list of goals that the automation hopes to achieve some time in the future: not a log
- ✗ D. A type of agent: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S100-M-Q2 · question page 2082

AI.AGENT.06 · Use logs to check automations and prevent duplicate actions

Why do automations retry steps?

Answer: C. Because a step may have failed

Recover from failures.

- ✗ A. To use more memory: no
- ✗ B. To slow down: no
- ✗ D. Because every step must be done exactly twice to make sure that it worked correctly: not the reason

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S100-M-Q3 · question page 2083

AI.AGENT.06 · Use logs to check automations and prevent duplicate actions

In this prepared example, what went wrong?

Prepared example (a fictional automation log)

Time	Action	Order ID	Result
10:00:01	Charge 25 dollars	A-118	Timed out
10:00:31	Charge 25 dollars (retry)	A-118	Success
10:05:00	Bank statement check	A-118	2 charges found

Answer: D. The customer was charged twice

The timed-out charge actually went through.

- ✗ A. The order ID changed: same ID
- ✗ B. The customer was never charged at all, because the first attempt timed out completely: two charges found
- ✗ C. Nothing went wrong: double charge

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S100-M-Q4 · question page 2084

AI.AGENT.06 · Use logs to check automations and prevent duplicate actions

In the same example, how much was charged in total?

Prepared example (a fictional automation log)

Time	Action	Order ID	Result
10:00:01	Charge 25 dollars	A-118	Timed out
10:00:31	Charge 25 dollars (retry)	A-118	Success
10:05:00	Bank statement check	A-118	2 charges found

Answer: B. 50 dollars

25 + 25 = 50.

- ✗ A. 75 dollars: two charges
- ✗ C. 0 dollars: two charges
- ✗ D. 25 dollars: it did

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: none (stable)

S100-M-Q5 · question page 2085

AI.AGENT.06 · Use logs to check automations and prevent duplicate actions

How can a safe design prevent duplicates?

Answer: D. Check a unique ID before repeating

Each action once.

- ✗ A. Never retry anything: too strict
- ✗ B. Retry every step as many times as possible, so that at least one attempt is sure to work: more duplicates
- ✗ C. Delete the log: loses evidence

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S100-B1-Q1 · question page 2086

AI.AGENT.06 · Compare retries with pressing an elevator button twice

Why is pressing the elevator button twice harmless?

Answer: A. The elevator still comes once

Repeating does not change the result.

- ✗ B. Because buttons break: no
- ✗ C. Because elevators are slow: not the reason
- ✗ D. Because pressing a button twice always cancels the first press, so nothing happens: not how it works

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S100-B1-Q2 · question page 2087

AI.AGENT.06 · Compare retries with pressing an elevator button twice

Why can pressing "BUY" twice be a problem?

Answer: C. It might buy two items

Repeating changes the result.

✗ A. Because buttons are red: no

✗ B. Because pressing buy twice always cancels the order completely and refunds your money: not reliable

✗ D. Because the page gets brighter: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S100-B1-Q3 · question page 2088

AI.AGENT.06 · Compare retries with pressing an elevator button twice

A retried "send payment" step is like what?

Answer: B. Pressing "BUY" twice

Risk of duplicates.

✗ A. Pressing the elevator button twice, which is always harmless for everyone involved: it is not harmless

✗ C. Reading a menu: no

✗ D. Closing a door: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S100-B1-Q4 · question page 2089

AI.AGENT.06 · Compare retries with pressing an elevator button twice

What should the automation check before retrying a buy-like step?

Answer: A. Whether the first attempt worked

Use a record.

✗ B. The weather: no

✗ C. The screen color: no

✗ D. Whether the internet is fast enough: not the key

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S100-B1-Q5 · question page 2090

AI.AGENT.06 · Compare retries with pressing an elevator button twice

What record can show if the purchase happened?

Answer: B. An order number

A unique ID.

- ✗ A. The time of day: not enough
- ✗ C. The browser name: no
- ✗ D. The color of the "BUY" button on the website, which changes after a purchase: not reliable

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S100-F1-Q1 · question page 2091

AI.AGENT.06 · Use logs and safe retries in practice

A parent says they got the same reminder text three times. Where should you look first?

Answer: C. The automation's log

Logs show what happened.

- ✗ A. The parent's phone settings, since repeated texts are always caused by phones: check the system
- ✗ B. The school website: not the record
- ✗ D. Nowhere: investigate

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S100-F1-Q2 · question page 2092

AI.AGENT.06 · Use logs and safe retries in practice

A log shows "send reminder" retried after a timeout, and both attempts succeeded. What happened?

Answer: D. A duplicate message was sent

Retries can duplicate.

- ✗ A. No message was sent: two succeeded
- ✗ B. The log is wrong: not shown
- ✗ C. Nothing unusual, since the second attempt cancels the first message automatically: not reliable

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S100-F1-Q3 · question page 2093

AI.AGENT.06 · Use logs and safe retries in practice

Which detail makes a retry safe for payments?

Answer: C. A unique payment ID checked before charging

Detect duplicates.

- ✗ A. A new card each time: no
- ✗ B. A bigger payment: no
- ✗ D. A shorter waiting time between attempts: not prevention

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S100-F1-Q4 · question page 2094

AI.AGENT.06 · Use logs and safe retries in practice

Which log entry is most useful?

Answer: A. "9:02 sent reminder to Sam - success"

What, when, who, result.

- ✗ B. "OK": vague
- ✗ C. "Something happened at some point today," which shows that the system is working: vague

✗ D. "Done": vague

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S100-F1-Q5 · question page 2095

AI.AGENT.06 · Use logs and safe retries in practice

An automation never retries failed steps. What is the downside?

Answer: D. Real failures stay unfixed

Retries have value.

✗ A. It runs too fast: not the issue
✗ B. It sends duplicates: opposite
✗ C. There is no downside: not always

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S100-F2-Q1 · question page 2096

AI.AGENT.06 · Design idempotent steps and useful logs

Which step is idempotent?

Answer: B. "Set status to paid"

Repeating has the same effect.

✗ A. "Add 10 dollars to the balance," since adding money twice is the same as adding it once: not idempotent
✗ C. "Charge the card": duplicates
✗ D. "Send a message": duplicates

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S100-F2-Q2 · question page 2097

AI.AGENT.06 · Design idempotent steps and useful logs

"Add 10 dollars" runs twice. What is the result?

Answer: A. 20 dollars added

10 + 10 = 20.

- ✗ B. 0 dollars added: no
- ✗ C. 30 dollars added: no
- ✗ D. 10 dollars added: not idempotent

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: none (stable)

S100-F2-Q3 · question page 2098

AI.AGENT.06 · Design idempotent steps and useful logs

What should logs avoid storing?

Answer: C. Secrets such as passwords

Keep secrets out of logs.

- ✗ A. The result of each action: needed
- ✗ B. Which action ran: needed
- ✗ D. The time of each action: needed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S100-F2-Q4 · question page 2099

AI.AGENT.06 · Design idempotent steps and useful logs

Why spot-check logs even when things seem fine?

Answer: B. To catch quiet problems early

Problems can be silent.

- ✗ A. Because logs are deleted automatically if nobody reads them for more than one day: not the reason

✗ C. Because logs are always wrong: no

✗ D. To make the logs longer: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S100-F2-Q5 · question page 2100

AI.AGENT.06 · Design idempotent steps and useful logs

What should a useful log entry include?

Answer: D. What, when, who and the result

Enough to rebuild events.

✗ A. Only the date: not enough

✗ B. Only a smiley face to show that the step finished, which is easy to read quickly: not enough

✗ C. Only the password used: never

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S101-M-Q1 · question page 2102

AI.WORK.01 · Analyze how AI may change some tasks in a job

What is a job made of?

Answer: C. Many tasks

Tasks make up jobs.

✗ A. Only a job title: no

✗ B. Only a salary: no

✗ D. One single task: usually many

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S101-M-Q2 · question page 2103

AI.WORK.01 · Analyze how AI may change some tasks in a job

In this prepared example, which task is least likely to change?

Prepared example (an illustrative task breakdown, not a forecast)

Task (medical receptionist)	How AI might affect it
Draft appointment reminders	Could help a lot
Fill insurance forms	Could help, with checking
Comfort a worried patient	Little change
Solve a scheduling surprise	Some help; judgment needed

Answer: B. Comfort a worried patient

Human care and trust.

- ✗ A. Fill insurance forms: could help
- ✗ C. Solve a scheduling surprise: some help
- ✗ D. Draft appointment reminders: could help a lot

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S101-M-Q3 · question page 2104

AI.WORK.01 · Analyze how AI may change some tasks in a job

Why is "Will AI take this job?" usually the wrong question?

Answer: D. AI tends to change some tasks, not all

Think in tasks.

- ✗ A. Because nobody knows what jobs are: no
- ✗ B. Because jobs have no tasks: they do
- ✗ C. Because AI will never affect any job: too strong

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S101-M-Q4 · question page 2105

AI.WORK.01 · Analyze how AI may change some tasks in a job

Which kind of task depends most on a person?

Answer: A. One needing trust and responsibility

Judgment, trust, presence.

- ✗ B. Drafting reminders: routine
- ✗ C. Sorting files: routine
- ✗ D. One that involves copying the same information from one form into another form: routine

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S101-M-Q5 · question page 2106

AI.WORK.01 · Analyze how AI may change some tasks in a job

What new task might AI create in a job?

Answer: C. Checking AI drafts

New tasks appear.

- ✗ A. No new tasks at all, since AI only ever removes tasks and never adds any work: it can add
- ✗ B. Only physical tasks: no
- ✗ D. Only unpaid tasks: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S101-B1-Q1 · question page 2107

AI.WORK.01 · Compare a job with a toolbox of tasks

What does the power drill replace?

Answer: A. Some uses of the hand screwdriver

Some tasks change.

- ✗ B. The carpenter completely, since a drill can build an entire house without a person: no
- ✗ C. The hammer: no
- ✗ D. The tape measure: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S101-B1-Q2 · question page 2108

AI.WORK.01 · Compare a job with a toolbox of tasks

What is still needed?

Answer: B. The carpenter's judgment and other tools

Other tasks remain.

- ✗ A. Only the box: no
- ✗ C. Only the drill: no
- ✗ D. Nothing else, because the drill does every task that the carpenter used to do: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S101-B1-Q3 · question page 2109

AI.WORK.01 · Compare a job with a toolbox of tasks

What new task appears?

Answer: D. Using the drill safely

New skills.

- ✗ A. Painting the drill: no
- ✗ B. Selling the house: no

✗ C. Throwing away every other tool in the box, since the drill makes them unnecessary: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S101-B1-Q4 · question page 2110

AI.WORK.01 · Compare a job with a toolbox of tasks

Did the carpenter's job vanish?

Answer: C. No; it changed

Change, not disappearance.

✗ A. Yes, partly: it changed
✗ B. It cannot be known: the example shows
✗ D. Yes: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S101-B1-Q5 · question page 2111

AI.WORK.01 · Compare a job with a toolbox of tasks

In this comparison, AI is like what?

Answer: D. A new tool in the box

One tool among many.

✗ A. The wood: no
✗ B. The carpenter: no
✗ C. The whole toolbox: too broad

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S101-F1-Q1 · question page 2112

AI.WORK.01 · Break jobs into tasks in practice

A bank teller counts cash, answers questions, spots fraud and helps elderly customers. Which task might AI help with most?

Answer: A. Spotting unusual transactions

Pattern finding.

- ✗ B. Opening the door: physical
- ✗ C. Helping elderly customers feel comfortable and safe while they visit the bank in person: human trust
- ✗ D. Shaking hands: physical

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S101-F1-Q2 · question page 2113

AI.WORK.01 · Break jobs into tasks in practice

A teacher plans lessons, grades quizzes, mentors students and handles conflicts. Which is least likely to change?

Answer: B. Mentoring students

Relationship and judgment.

- ✗ A. Typing announcements: routine
- ✗ C. Grading multiple-choice quizzes, since grading needs a teacher's personal judgment each time: routine
- ✗ D. Formatting worksheets: routine

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S101-F1-Q3 · question page 2114

AI.WORK.01 · Break jobs into tasks in practice

Which is the better question about a graphic designer's job?

Answer: A. "Which design tasks might AI change?"

Task-level question.

- ✗ B. "Is design a real job?": not helpful
- ✗ C. "Will AI completely replace every graphic designer in the world by next year?": all-or-nothing
- ✗ D. "Do designers use computers?": not helpful

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S101-F1-Q4 · question page 2115

AI.WORK.01 · Break jobs into tasks in practice

A warehouse adds AI scheduling. What new task might workers gain?

Answer: B. Checking and adjusting schedules

Oversight tasks.

- ✗ A. None, because adding a scheduling tool never creates any new work for anybody: it can
- ✗ C. Selling the software: no
- ✗ D. Building the warehouse: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S101-F1-Q5 · question page 2116

AI.WORK.01 · Break jobs into tasks in practice

Which task depends on physical presence?

Answer: C. Repairing a leaking pipe

Hands-on work.

- ✗ A. Translating a menu: not physical
- ✗ B. Summarizing a report: not physical
- ✗ D. Drafting a reply: not physical

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S101-F2-Q1 · question page 2117

AI.WORK.01 · Avoid simple predictions about jobs

Tasks in a job could be automated, but the job changes slowly. What might explain this?

Answer: D. Cost, rules or customer preferences

Many factors slow change.

- ✗ A. The weather: no
- ✗ B. Workers refusing to learn: too simple
- ✗ C. Nothing explains it, since automation always happens immediately once it is possible: not true

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S101-F2-Q2 · question page 2118

AI.WORK.01 · Avoid simple predictions about jobs

Which prediction is most reasonable?

Answer: C. "Some tasks will change; how much is uncertain."

Uncertain, task-level.

- ✗ A. "This whole job will certainly disappear within one year, everywhere in the world.": too certain
- ✗ B. "Nothing about this job will ever change.": too certain
- ✗ D. "AI will do every job by next month.": too certain

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S101-F2-Q3 · question page 2119

AI.WORK.01 · Avoid simple predictions about jobs

A job shrinks in one city but grows in another. What does this show?

Answer: D. Effects can differ by place

Uneven effects.

- ✗ A. That AI is not involved: not shown
- ✗ B. That one of the cities must be reporting completely false numbers about its jobs: not implied
- ✗ C. That jobs never change: they changed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S101-F2-Q4 · question page 2120

AI.WORK.01 · Avoid simple predictions about jobs

How can workers influence how their jobs change?

Answer: B. By learning to use and check new tools well

Active adaptation.

- ✗ A. By hiding from managers: no
- ✗ C. By changing their job title only: no
- ✗ D. By ignoring every new tool and hoping that nothing about their work ever changes: less influence

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S101-F2-Q5 · question page 2121

AI.WORK.01 · Avoid simple predictions about jobs

Why might a job need a person even if tasks could be automated?

Answer: A. Someone must be responsible

Accountability.

✗ B. Because people type faster: not the reason

✗ C. Because computers are not allowed to do any work at all on weekends or holidays: no

✗ D. Because automation is illegal: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S102-M-Q1 · question page 2123

AI.WORK.02 · Read an employment forecast as a projection with assumptions

What is an employment forecast?

Answer: A. An estimate based on assumptions

Projection, not promise.

✗ B. A list of job openings: no

✗ C. A salary table: no

✗ D. A guarantee that a certain number of jobs will exist, which employers must legally meet: not a guarantee

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: work_wef; work_bls_ds

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S102-M-Q2 · question page 2124

AI.WORK.02 · Read an employment forecast as a projection with assumptions

What was the WEF Future of Jobs Report 2025 based on?

Answer: D. A survey of more than 1,000 employers

Survey-based.

✗ A. A guess by one person: no

✗ B. One company's records: no

✗ C. A count of every single job that exists: a survey

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: work_wef; work_bls_ds

S102-M-Q3 · question page 2125

AI.WORK.02 · Read an employment forecast as a projection with assumptions

What net change did that report project by 2030?

Answer: B. 78 million roles

170 - 92 = 78.

- ✗ A. 92 million roles: that is displaced
- ✗ C. 262 million roles: not net
- ✗ D. 170 million roles: that is created

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: work_wef; work_bls_ds

S102-M-Q4 · question page 2126

AI.WORK.02 · Read an employment forecast as a projection with assumptions

Which question helps you read a forecast?

Answer: C. "What period and place does it cover?"

Scope matters.

- ✗ A. "Is it long?": irrelevant
- ✗ B. "Is it popular online?": irrelevant
- ✗ D. "Does the report have a colorful cover?": irrelevant

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: work_wef; work_bls_ds

S102-M-Q5 · question page 2127

AI.WORK.02 · Read an employment forecast as a projection with assumptions

Different forecasts disagree. What does this show?

Answer: A. Forecasts depend on assumptions and methods

Uncertainty.

✗ B. That forecasting is illegal: no

✗ C. That jobs cannot change: no

✗ D. That every forecast except the most recent one must be completely wrong and useless: not implied

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: work_wef; work_bls_ds

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S102-B1-Q1 · question page 2128

AI.WORK.02 · Compare a jobs forecast with a weather forecast

What does "70% chance of rain" mean?

Answer: B. Rain is likely but not certain

Probability, not promise.

✗ A. It will certainly rain: not certain

✗ C. It will not rain: likely rain

✗ D. It will rain for exactly 70%: not the meaning

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: work_wef; work_bls_ds

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S102-B1-Q2 · question page 2129

AI.WORK.02 · Compare a jobs forecast with a weather forecast

What is a jobs forecast based on?

Answer: D. Assumptions and methods

Like weather models.

✗ A. Certain knowledge of the future, which is why jobs forecasts are never wrong: no

✗ B. Guesses only: it has methods

✗ C. Weather data: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: work_wef; work_bls_ds

S102-B1-Q3 · question page 2130

AI.WORK.02 · Compare a jobs forecast with a weather forecast

Weather services sometimes disagree. What is the jobs parallel?

Answer: C. Different forecasts disagree

Different methods.

- ✗ A. Forecasts are useless: too strong
- ✗ B. All job forecasts always give exactly the same numbers for every country and every year: no
- ✗ D. Jobs never change: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: work_wef; work_bls_ds

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S102-B1-Q4 · question page 2131

AI.WORK.02 · Compare a jobs forecast with a weather forecast

Using a forecast to bring an umbrella is like what?

Answer: A. Using a jobs forecast to plan learning

Informed planning.

- ✗ B. Buying a weather app: no
- ✗ C. Ignoring all forecasts: not using them
- ✗ D. Quitting your current job immediately: overreaction

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: work_wef; work_bls_ds

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S102-B1-Q5 · question page 2132

AI.WORK.02 · Compare a jobs forecast with a weather forecast

What is the lesson?

Answer: B. Use forecasts as inputs, not guarantees

Useful but uncertain.

- ✗ A. Never read forecasts: too strong
- ✗ C. Only read weather forecasts: no
- ✗ D. Treat every forecast as a certain prediction: not certain

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: work_wef; work_bls_ds

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S102-F1-Q1 · question page 2133

AI.WORK.02 · Read job forecasts in practice

A headline says "AI will create 170 million jobs!" What is missing?

Answer: D. The roles displaced and the net change

Created minus displaced.

- ✗ A. The author's photo: irrelevant
- ✗ B. The report's font: irrelevant
- ✗ C. Nothing, since the number of jobs created is the only number that matters in any forecast: losses matter

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: work_wef; work_bls_ds

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S102-F1-Q2 · question page 2134

AI.WORK.02 · Read job forecasts in practice

A friend says the WEF report "proves" there will be 78 million more jobs. What is more accurate?

Answer: C. It projects that, based on an employer survey

A projection, not proof.

- ✗ A. It says jobs will fall: it projects a net gain
- ✗ B. It proves it, because reports from large organizations are always exactly correct about the future: projection
- ✗ D. It says nothing about jobs: it does

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: work_wef; work_bls_ds

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S102-F1-Q3 · question page 2135

AI.WORK.02 · Read job forecasts in practice

Two forecasts give very different numbers for the same field. What should you do?

Answer: B. Compare their methods and assumptions

Understand differences.

- ✗ A. Choose whichever forecast has the bigger number, since bigger numbers are more reliable: not a method
- ✗ C. Ignore both: too strong
- ✗ D. Average them blindly: not careful

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: work_wef; work_bls_ds

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S102-F1-Q4 · question page 2136

AI.WORK.02 · Read job forecasts in practice

Which detail is most important when reading a forecast?

Answer: D. Who made it and how

Source and method.

- ✗ A. Its page count: irrelevant
- ✗ B. Its color scheme: irrelevant
- ✗ C. How many times it has been shared on social media by people who liked it: popularity

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: work_wef; work_bls_ds

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S102-F1-Q5 · question page 2137

AI.WORK.02 · Read job forecasts in practice

A forecast covers 2025 to 2030 worldwide. Can it tell you about your town next year?

Answer: C. Not directly

Check scope.

- ✗ A. Yes, if the town is large: still differs
- ✗ B. Only for farming: no
- ✗ D. Yes, exactly, because worldwide numbers apply equally to every town in every year: scope differs

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: work_wef; work_bls_ds

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S102-F2-Q1 · question page 2138

AI.WORK.02 · Look beneath large numbers

A global net gain is projected. Could some workers still lose out?

Answer: A. Yes; losses can be hidden in a net figure

Net hides losses.

- ✗ B. Only if the gain is small: not the point
- ✗ C. Only in summer: no
- ✗ D. No, because a net gain means that every single worker in the world will be better off: not true

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: work_wef; work_bls_ds

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S102-F2-Q2 · question page 2139

AI.WORK.02 · Look beneath large numbers

A role grows "200%" from 5 jobs to 15. How many jobs were added?

Answer: B. 10

15 - 5 = 10.

✗ A. 15: that is the total

✗ C. 5: that is the start

✗ D. 200: percent is not count

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: work_wef; work_bls_ds

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S102-F2-Q3 · question page 2140

AI.WORK.02 · Look beneath large numbers

BLS projects data scientist jobs to grow 34% from 2024 to 2034. What is this?

Answer: A. A US projection with its own method

Government projection.

✗ B. A worldwide count: US only

✗ C. A guarantee that every person who studies data science will get a job by 2034: not a guarantee

✗ D. A salary promise: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: work_wef; work_bls_ds

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S102-F2-Q4 · question page 2141

AI.WORK.02 · Look beneath large numbers

Why compare like with like?

Answer: D. Forecasts differ in place, period and what they count

Matching scope.

✗ A. Because every forecast always measures exactly the same thing in exactly the same way: they differ

✗ B. Because older forecasts are always wrong: no

✗ C. Because numbers must be the same: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: work_wef; work_bls_ds

S102-F2-Q5 · question page 2142

AI.WORK.02 · Look beneath large numbers

How should you use forecasts in decisions?

Answer: C. As one input, not a guarantee

Weigh with other information.

- ✗ A. Not at all: too strong
- ✗ B. Only when they agree with you: biased
- ✗ D. As the only input: too much weight

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: work_wef; work_bls_ds

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S103-M-Q1 · question page 2144

AI.WORK.03 · Describe what technical AI roles typically do

What does a data scientist typically do?

Answer: B. Collects, cleans and analyzes data to answer questions

Data to answers.

- ✗ A. Builds the physical computer chips that every AI model needs in order to run at all: not the role
- ✗ C. Sells AI products: not the role
- ✗ D. Writes laws about AI: not the role

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: work_bls_ds

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S103-M-Q2 · question page 2145

AI.WORK.03 · Describe what technical AI roles typically do

What does an ML engineer typically do?

Answer: C. Builds and runs models so they work reliably in products

Models in production.

- ✗ A. Only answers customer emails about products that use machine learning features: not the role
- ✗ B. Designs logos: no
- ✗ D. Repairs printers: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: work_bls_ds

---

S103-M-Q3 · question page 2146

AI.WORK.03 · Describe what technical AI roles typically do

What does an AI engineer often do?

Answer: A. Builds applications on top of existing models

Applications using models.

- ✗ B. Manages office supplies: no
- ✗ C. Teaches only history: no
- ✗ D. Trains every model from nothing, using only paper and pencil calculations each day: not typical

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: work_bls_ds

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S103-M-Q4 · question page 2147

AI.WORK.03 · Describe what technical AI roles typically do

What skills do all three roles usually need?

Answer: D. Programming, testing and communication

Shared skills.

- ✗ A. No skills: no
- ✗ B. Only drawing: no
- ✗ C. Only a famous university degree, since skills are not important for

technical roles: not stated

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: work_bls_ds

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S103-M-Q5 · question page 2148

AI.WORK.03 · Describe what technical AI roles typically do

Why does this course not state pay or credentials?

Answer: A. They vary widely by employer and place

Too variable to state.

- ✗ B. Because all jobs pay the same: no
- ✗ C. Because pay is secret everywhere: not the reason
- ✗ D. Because these roles are always unpaid: not true

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: work_bls_ds

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S103-B1-Q1 · question page 2149

AI.WORK.03 · Compare AI roles with roles in a restaurant

The food taster who studies sales is like whom?

Answer: D. A data scientist

Analyzing data.

- ✗ A. An AI engineer: no
- ✗ B. A cashier: no
- ✗ C. An ML engineer: not the comparison

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: work_bls_ds

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S103-B1-Q2 · question page 2150

AI.WORK.03 · Compare AI roles with roles in a restaurant

The kitchen engineer is like whom?

Answer: B. An ML engineer

Reliable systems.

- ✗ A. A waiter: no
- ✗ C. A chef: no
- ✗ D. A data scientist: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: work_bls_ds

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S103-B1-Q3 · question page 2151

AI.WORK.03 · Compare AI roles with roles in a restaurant

The chef using ready-made ingredients is like whom?

Answer: C. An AI engineer

Building on existing parts.

- ✗ A. A dishwasher: no
- ✗ B. An ML engineer: no
- ✗ D. A data scientist: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: work_bls_ds

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S103-B1-Q4 · question page 2152

AI.WORK.03 · Compare AI roles with roles in a restaurant

What does the kitchen engineer care about most?

Answer: B. Food coming out right every time

Reliability.

- ✗ A. Choosing music: no
- ✗ C. Counting customers: that is analysis

✗ D. Inventing a brand-new dish: that is the chef

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: work_bls_ds

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S103-B1-Q5 · question page 2153

AI.WORK.03 · Compare AI roles with roles in a restaurant

What do all three restaurant roles share?

Answer: A. Working together to serve customers well

Different tasks, shared goal.

✗ B. Only the uniform: no

✗ C. Exactly the same daily tasks, done in the same way, with the same tools and training: they differ

✗ D. Nothing: they share goals

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: work_bls_ds

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S103-F1-Q1 · question page 2154

AI.WORK.03 · Match tasks to technical roles in practice

Someone analyzes survey results to find why customers leave. Which role fits best?

Answer: D. Data scientist

Analysis to answer questions.

✗ A. Graphic designer: no

✗ B. Receptionist: no

✗ C. ML engineer: not needed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: work_bls_ds

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S103-F1-Q2 · question page 2155

AI.WORK.03 · Match tasks to technical roles in practice

Someone makes sure a recommendation model keeps working when millions use it. Which role fits?

Answer: C. ML engineer

Reliable model systems.

- ✗ A. Accountant: no
- ✗ B. Copywriter: no
- ✗ D. Data scientist: different focus

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: work_bls_ds

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S103-F1-Q3 · question page 2156

AI.WORK.03 · Match tasks to technical roles in practice

Someone builds a help-desk app that uses an existing language model and company documents. Which role fits?

Answer: A. AI engineer

Application on a model.

- ✗ B. Hardware technician: no
- ✗ C. Salesperson: no
- ✗ D. Data scientist: not typical

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: work_bls_ds

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S103-F1-Q4 · question page 2157

AI.WORK.03 · Match tasks to technical roles in practice

A posting titled "AI Specialist" lists data cleaning, statistics and reports. What role is it most like?

Answer: B. Data scientist

Tasks reveal the role.

- ✗ A. Pilot: no

✗ C. Chef: no

✗ D. ML engineer: read tasks

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: work_bls_ds

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S103-F1-Q5 · question page 2158

AI.WORK.03 · Match tasks to technical roles in practice

Which statement about these roles is accurate?

Answer: C. Titles and requirements vary by employer

Variation is normal.

✗ A. Only one role exists: several

✗ B. No programming is ever needed: usually needed

✗ D. Every employer uses exactly the same title: they vary

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: work_bls_ds

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S103-F2-Q1 · question page 2159

AI.WORK.03 · Read job postings and titles carefully

Why read tasks instead of titles in a posting?

Answer: D. Titles overlap and differ between employers

Tasks are clearer.

✗ A. Because titles are always false and are written only to attract as many applicants as possible: too strong

✗ B. Because tasks are shorter: not the reason

✗ C. Because titles are secret: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: work_bls_ds

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S103-F2-Q2 · question page 2160

AI.WORK.03 · Read job postings and titles carefully

A posting lists 25 skills. What is a reasonable reading?

Answer: B. It describes an ideal, not a minimum

Wish lists are common.

- ✗ A. The job needs no skills: no
- ✗ C. The job is fake: not implied
- ✗ D. Every applicant must have all 25 skills: usually not

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: work_bls_ds

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S103-F2-Q3 · question page 2161

AI.WORK.03 · Read job postings and titles carefully

Which source describes typical duties of US occupations consistently?

Answer: C. The BLS Occupational Outlook Handbook

Government reference.

- ✗ A. A movie: no
- ✗ B. A random job posting: one employer
- ✗ D. A single social media post from someone: not consistent

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: work_bls_ds

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S103-F2-Q4 · question page 2162

AI.WORK.03 · Read job postings and titles carefully

Two companies call different jobs "data scientist." What does this show?

Answer: A. Titles are not standard

Compare tasks.

- ✗ B. That data science is not real: no

✗ C. That one of the two companies must be breaking the law by using the wrong job title: no

✗ D. That titles never matter: too strong

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: work_bls_ds

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S103-F2-Q5 · question page 2163

AI.WORK.03 · Read job postings and titles carefully

A posting says "ML engineer" but describes only writing reports. What should you conclude?

Answer: D. The tasks matter more than the title

Read the tasks.

✗ A. That ML engineers only write reports: no

✗ B. That you should ignore the tasks: no

✗ C. That the posting is definitely a scam: not necessarily

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: work_bls_ds

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S104-M-Q1 · question page 2165

AI.WORK.04 · Describe AI-related roles beyond coding

What does an AI evaluator do?

Answer: B. Designs tests and judges output quality and safety

Testing and judging.

✗ A. Cleans offices: no

✗ C. Sells computers: no

✗ D. Writes every line of code for a new model from scratch: not evaluation

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S104-M-Q2 · question page 2166

AI.WORK.04 · Describe AI-related roles beyond coding

What do governance and policy roles do?

Answer: A. Set rules and check compliance

Rules and oversight.

- ✗ B. Train models: technical
- ✗ C. Label images: data work
- ✗ D. Design the colors and icons: that is design

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S104-M-Q3 · question page 2167

AI.WORK.04 · Describe AI-related roles beyond coding

Why are subject experts valuable in AI work?

Answer: D. They judge whether output makes sense in their field

Domain knowledge.

- ✗ A. Because they never need AI: not the point
- ✗ B. Because they work for free: no
- ✗ C. Because they can write code faster than any programmer: not the reason

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S104-M-Q4 · question page 2168

AI.WORK.04 · Describe AI-related roles beyond coding

What does a product manager decide?

Answer: C. What a tool should do and for whom

Purpose and users.

✗ A. Which laws apply: legal

✗ B. How to repair servers: no

✗ D. Which data labels are correct: data work

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S104-M-Q5 · question page 2169

AI.WORK.04 · Describe AI-related roles beyond coding

Which skill is valuable across all these roles?

Answer: B. Careful checking plus domain knowledge

Judgment in context.

✗ A. Memorizing model names: not key

✗ C. Owning many devices: no

✗ D. Typing very fast, which matters more than understanding the work in any role: not key

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S104-B1-Q1 · question page 2170

AI.WORK.04 · Compare AI roles with a film crew

Film editors who cut what does not work are like whom?

Answer: A. Evaluators

Reviewing quality.

✗ B. Coders: no

✗ C. Lawyers: no

✗ D. Product managers: that is writers

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S104-B1-Q2 · question page 2171

AI.WORK.04 · Compare AI roles with a film crew

Writers who decide the story are like whom?

Answer: C. Product managers

Deciding purpose.

- ✗ A. Subject experts: no
- ✗ B. Lawyers: no
- ✗ D. Evaluators: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S104-B1-Q3 · question page 2172

AI.WORK.04 · Compare AI roles with a film crew

Research staff checking history are like whom?

Answer: D. Subject experts

Domain knowledge.

- ✗ A. Designers: no
- ✗ B. Coders: no
- ✗ C. Lawyers: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S104-B1-Q4 · question page 2173

AI.WORK.04 · Compare AI roles with a film crew

Lawyers clearing rights are like whom?

Answer: C. Governance roles

Rules and compliance.

✗ A. Camera operators: no

✗ B. Writers: no

✗ D. Evaluators: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S104-B1-Q5 · question page 2174

AI.WORK.04 · Compare AI roles with a film crew

What is the lesson?

Answer: D. Coding is one part of a larger team

Many roles.

✗ A. Films do not need editors: no

✗ B. AI has no teams: no

✗ C. Only people who write code can ever work on AI projects in any company: not true

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S104-F1-Q1 · question page 2175

AI.WORK.04 · Identify non-coding AI roles in practice

A nurse reviews AI summaries of patient charts for errors. What role is this most like?

Answer: A. Subject expert evaluator

Domain judgment.

✗ B. ML engineer, because reviewing summaries requires building and training the model: no

✗ C. Lawyer: no

✗ D. Product designer: not mainly

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S104-F1-Q2 · question page 2176

AI.WORK.04 · Identify non-coding AI roles in practice

Someone writes company rules for which AI tools staff may use. What role is this?

Answer: B. Governance or policy

Rules.

- ✗ A. Coding: no
- ✗ C. Evaluation: not mainly
- ✗ D. Data labeling, since writing rules is the same as tagging images for training data: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S104-F1-Q3 · question page 2177

AI.WORK.04 · Identify non-coding AI roles in practice

Someone decides an AI homework helper should focus on hints, not answers. What role is this?

Answer: A. Product management

What the tool does.

- ✗ B. Hardware: no
- ✗ C. Legal: no
- ✗ D. Data science: not mainly

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S104-F1-Q4 · question page 2178

AI.WORK.04 · Identify non-coding AI roles in practice

Someone tags thousands of photos as "cat" or "not cat." What kind of work is this?

Answer: C. Data labeling

Data work.

- ✗ A. Design: no
- ✗ B. Governance: no
- ✗ D. Product management: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S104-F1-Q5 · question page 2179

AI.WORK.04 · Identify non-coding AI roles in practice

An accountant wants to work with AI without becoming a programmer. What is a strong route?

Answer: B. Combine accounting knowledge with AI understanding

Field plus AI literacy.

- ✗ A. Only learn model names: not enough
- ✗ C. Avoid AI entirely: misses opportunity
- ✗ D. Abandon accounting completely and start again: often weaker

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S104-F2-Q1 · question page 2180

AI.WORK.04 · Weigh technical literacy and job conditions

Why do non-coding AI roles still need technical literacy?

Answer: D. To understand how AI works and fails

Know capabilities and limits.

- ✗ A. To memorize jargon: not the point
- ✗ B. Because every non-coding role secretly requires writing large amounts of code each day: not true
- ✗ C. To repair computers: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not

applicable. Sources: none (stable)

S104-F2-Q2 · question page 2181

AI.WORK.04 · Weigh technical literacy and job conditions

A nurse explains why an AI chart summary is unsafe. Why is this valuable?

Answer: B. Domain expertise catches risks

Expert judgment.

- ✗ A. Because nurses write code: not the point
- ✗ C. Because it is fast: not the point
- ✗ D. It is not valuable: not true

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S104-F2-Q3 · question page 2182

AI.WORK.04 · Weigh technical literacy and job conditions

Some data-labeling jobs are low-paid and repetitive. What should a job seeker do?

Answer: C. Look at conditions as well as titles

Check conditions.

- ✗ A. Avoid all AI jobs: too strong
- ✗ B. Ignore pay: not wise
- ✗ D. Assume all AI jobs pay very well: not true

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S104-F2-Q4 · question page 2183

AI.WORK.04 · Weigh technical literacy and job conditions

Which skill helps across non-coding AI roles?

Answer: D. Describing problems precisely

Precise communication.

- ✗ A. Typing only in capitals: no
- ✗ B. Using as much jargon as possible: unclear
- ✗ C. Avoiding any discussion of problems: unhelpful

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S104-F2-Q5 · question page 2184

AI.WORK.04 · Weigh technical literacy and job conditions

What is often stronger than switching fields entirely?

Answer: A. Combining your field with AI understanding

Build on strengths.

- ✗ B. Changing job titles: not substance
- ✗ C. Staying unaware of AI: misses opportunity
- ✗ D. Learning only the newest AI tool names every week: shallow

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S105-M-Q1 · question page 2186

AI.WORK.05 · Plan a learning route and portfolio for a stated goal

What is a learning route?

Answer: A. A plan from where you are to a stated goal

Goal-driven plan.

- ✗ B. A map of a school: no
- ✗ C. A certificate: no
- ✗ D. A list of every AI course that exists: no goal

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S105-M-Q2 · question page 2187

AI.WORK.05 · Plan a learning route and portfolio for a stated goal

How should you build a route?

Answer: D. Work backward from the goal's tasks and skills

Goal, tasks, skills, resources.

- ✗ A. Copy a friend's plan exactly: may not fit
- ✗ B. Pick courses at random: no
- ✗ C. Start with the most expensive course available: not the method

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S105-M-Q3 · question page 2188

AI.WORK.05 · Plan a learning route and portfolio for a stated goal

What is a portfolio?

Answer: C. Evidence of what you can do

Projects and write-ups.

- ✗ A. A list of every course you signed up for, including ones you did not finish: not evidence
- ✗ B. A folder of certificates only: too narrow
- ✗ D. A résumé template: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S105-M-Q4 · question page 2189

AI.WORK.05 · Plan a learning route and portfolio for a stated goal

Does any course or portfolio guarantee a job?

Answer: B. No

Evidence helps; nothing guarantees.

- ✗ A. Yes, if it is expensive: no
- ✗ C. Yes, after one year: no
- ✗ D. Yes: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S105-M-Q5 · question page 2190

AI.WORK.05 · Plan a learning route and portfolio for a stated goal

What should portfolio pieces match?

Answer: A. The goal

Targeted evidence.

- ✗ B. Other people's portfolios: not the point
- ✗ C. The longest projects possible: not the point
- ✗ D. Whatever topic is currently most popular online, even if it has nothing to do with the goal: off target

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S105-B1-Q1 · question page 2191

AI.WORK.05 · Work backward from a goal

What is the first step?

Answer: B. State the goal clearly

Goal first.

- ✗ A. List skills: step 3
- ✗ C. Buy a course before deciding what you want to achieve with the new skills: goal first
- ✗ D. Make a project: step 4

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S105-B1-Q2 · question page 2192

AI.WORK.05 · Work backward from a goal

In the example, which is a task?

Answer: C. Check facts

Something done.

- ✗ A. The club: no
- ✗ B. A sample newsletter: evidence
- ✗ D. Privacy basics: a skill

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S105-B1-Q3 · question page 2193

AI.WORK.05 · Work backward from a goal

In the example, which is a skill?

Answer: D. Fact-checking

What a task needs.

- ✗ A. The goal: no
- ✗ B. Draft: task
- ✗ C. A sample newsletter: evidence

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S105-B1-Q4 · question page 2194

AI.WORK.05 · Work backward from a goal

In the example, what is the evidence?

Answer: B. A sample newsletter with notes

Shows the skills.

- ✗ A. A promise to the club that every future newsletter will be perfectly accurate: not evidence
- ✗ C. A list of tasks: not evidence
- ✗ D. A certificate: not this plan

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S105-B1-Q5 · question page 2195

AI.WORK.05 · Work backward from a goal

What order do the steps follow?

Answer: A. Goal, tasks, skills, evidence

Work backward.

- ✗ B. Skills, goal, evidence, tasks: no
- ✗ C. Any order: order matters
- ✗ D. Evidence, skills, tasks, goal: reversed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S105-F1-Q1 · question page 2196

AI.WORK.05 · Plan routes and portfolios in practice

Jay's goal is an entry-level data analysis role. Which portfolio piece fits best?

Answer: D. A project analyzing a public data set with a clear write-up

Matches the goal.

- ✗ A. A collection of AI-generated poems, which shows creativity with many different tools: off target
- ✗ B. A logo design: off target

✗ C. A list of courses started: not evidence

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S105-F1-Q2 · question page 2197

AI.WORK.05 · Plan routes and portfolios in practice

A program promises "guaranteed job placement in 30 days." What should you think?

Answer: C. Be skeptical; nothing guarantees a job

Doubt guarantees.

✗ A. Sign up immediately, since programs are only allowed to promise things they can deliver: not reliable

✗ B. Ask for a longer guarantee: no

✗ D. Promises are always true: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S105-F1-Q3 · question page 2198

AI.WORK.05 · Plan routes and portfolios in practice

A teacher wants to use AI to make quizzes safely. What is a useful first skill?

Answer: A. Checking generated questions for errors

Matches the task.

✗ B. Buying the newest computer: not a skill

✗ C. Learning every AI law worldwide: too broad

✗ D. Building a new language model from scratch: unneeded

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S105-F1-Q4 · question page 2199

AI.WORK.05 · Plan routes and portfolios in practice

Which project is best sized for a beginner's portfolio?

Answer: B. A small project that can be finished well

Finishable scope.

- ✗ A. Ten unfinished projects: weak evidence
- ✗ C. A huge project that rebuilds an entire search engine, even if it never gets finished: too large
- ✗ D. No project: no evidence

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S105-F1-Q5 · question page 2200

AI.WORK.05 · Plan routes and portfolios in practice

Which resource choice fits a learner on a budget?

Answer: D. Free or low-cost resources that teach the needed skills

Matched and affordable.

- ✗ A. The most expensive course available, because price is the best sign of quality in learning: not necessarily
- ✗ B. Random videos with no plan: unfocused
- ✗ C. No resources: no learning

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S105-F2-Q1 · question page 2201

AI.WORK.05 · Make portfolios honest and keep routes current

Should a portfolio say where AI help was used?

Answer: C. Yes; be honest about what you did

Transparency.

✗ A. Never: dishonest

✗ B. Only if asked twice: be proactive

✗ D. No, because mentioning AI help always makes employers think the work has no value at all: honesty matters

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S105-F2-Q2 · question page 2202

AI.WORK.05 · Make portfolios honest and keep routes current

Which write-up is strongest?

Answer: A. Problem, approach, checks, result and what did not work

Shows thinking.

✗ B. Only the final result, presented as perfect, with no mention of any problems along the way: less credible

✗ C. Only the tools used: incomplete

✗ D. Only the time spent: incomplete

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S105-F2-Q3 · question page 2203

AI.WORK.05 · Make portfolios honest and keep routes current

A learner finds a skill matters more than expected. What should they do?

Answer: B. Adjust the route

Update as you learn.

✗ A. Ignore the skill: no

✗ C. Keep the original plan unchanged, since changing plans always shows a lack of commitment: rigid

✗ D. Quit learning: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not

applicable. Sources: none (stable)

S105-F2-Q4 · question page 2204

AI.WORK.05 · Make portfolios honest and keep routes current

Who can tell you what evidence employers value?

Answer: D. People working in the field

Ask practitioners.

- ✗ A. Random strangers online only: unreliable
- ✗ B. No one: people can
- ✗ C. Advertisements for courses: biased

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S105-F2-Q5 · question page 2205

AI.WORK.05 · Make portfolios honest and keep routes current

Why include what did not work?

Answer: C. It shows honest problem-solving

Honest reflection.

- ✗ A. It should never be included: it helps
- ✗ B. To make the write-up longer: not the point
- ✗ D. Because employers only hire people whose projects failed completely and never worked: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S106-M-Q1 · question page 2207

AI.WORK.06 · Spot exaggerated AI income, job or course promises

Which is a warning sign in an AI offer?

Answer: D. Guaranteed income

Guarantees are red flags.

- ✗ A. A free trial lesson: not a warning sign
- ✗ B. Independent reviews: good sign
- ✗ C. A clear description: good sign

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_ftc_aicomply

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S106-M-Q2 · question page 2208

AI.WORK.06 · Spot exaggerated AI income, job or course promises

"Only 3 spots left - sign up in the next hour!" is what kind of sign?

Answer: C. Urgent pressure

Pressure to skip thinking.

- ✗ A. A refund policy: no
- ✗ B. An independent review: no
- ✗ D. A sign of quality: pressure tactic

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_ftc_aicomply

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S106-M-Q3 · question page 2209

AI.WORK.06 · Spot exaggerated AI income, job or course promises

What was "Operation AI Comply" (September 2024)?

Answer: A. FTC actions against deceptive AI claims

Consumer-protection enforcement.

- ✗ B. A coding contest: no
- ✗ C. A tech conference: no
- ✗ D. A new law that made all AI courses free: not what it was

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_ftc_aicomply

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S106-M-Q4 · question page 2210

AI.WORK.06 · Spot exaggerated AI income, job or course promises

Under what law did the FTC act?

Answer: B. Existing consumer-protection law

Existing rules apply to AI.

- ✗ A. A brand-new AI law that was passed by Congress specifically for that single operation: existing law
- ✗ C. Traffic law: no
- ✗ D. Tax law only: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_ftc_aicomply

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S106-M-Q5 · question page 2211

AI.WORK.06 · Spot exaggerated AI income, job or course promises

Testimonials you cannot verify are what?

Answer: A. A warning sign

Unverifiable praise.

- ✗ B. Strong proof that the offer works, since real people would not write fake reviews: not proof
- ✗ C. A legal requirement: no
- ✗ D. A refund: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_ftc_aicomply

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S106-B1-Q1 · question page 2212

AI.WORK.06 · Compare AI income claims with "get rich quick" ads in the past

What did old "envelope stuffing" ads promise?

Answer: B. Easy money with no skills

Too good to be true.

- ✗ A. A steady job with an established company, along with training, benefits and a contract: not what was promised
- ✗ C. Free envelopes: no
- ✗ D. A college degree: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_ftc_aicomply

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S106-B1-Q2 · question page 2213

AI.WORK.06 · Compare AI income claims with "get rich quick" ads in the past

What did buyers have to do first?

Answer: C. Pay a fee to learn the secret

Upfront fees.

- ✗ A. Move to a new city: no
- ✗ B. Nothing: they paid
- ✗ D. Pass a difficult test that proved they had years of experience in the same job: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_ftc_aicomply

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S106-B1-Q3 · question page 2214

AI.WORK.06 · Compare AI income claims with "get rich quick" ads in the past

What usually happened?

Answer: D. Most buyers made nothing

The promise failed.

✗ A. Every buyer became wealthy within a few months, exactly as the advertisements had promised: no

✗ B. They got refunds: not usually

✗ C. They got jobs: not usually

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_ftc_aicomply

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S106-B1-Q4 · question page 2215

AI.WORK.06 · Compare AI income claims with "get rich quick" ads in the past

How do AI income claims often compare?

Answer: C. Same pattern, new words

Old trick, new tool.

✗ A. They are always legal: not always

✗ B. They are always free: no

✗ D. They are completely different: same pattern

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_ftc_aicomply

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S106-B1-Q5 · question page 2216

AI.WORK.06 · Compare AI income claims with "get rich quick" ads in the past

What is the lesson?

Answer: D. Recognize the pattern whatever technology it mentions

Pattern recognition.

✗ A. Only trust envelopes: no

✗ B. Avoid all new technology: too strong

✗ C. Trust any offer that mentions AI, since AI is new and old scams do not apply to it: opposite

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_ftc_aicomply

S106-F1-Q1 · question page 2217

AI.WORK.06 · Check job and course claims in practice

An ad says "Earn 10,000 dollars a month with AI, no experience needed - pay 499 dollars to start." What should you do?

Answer: B. Treat it as a likely scam and research it

Several warning signs.

- ✗ A. Ask for a bigger guarantee: still a red flag
- ✗ C. Share it with friends: spreads it
- ✗ D. Pay quickly before the price goes up: red flags

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_ftc_aicomply

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S106-F1-Q2 · question page 2218

AI.WORK.06 · Check job and course claims in practice

A course says "100% job guarantee." What is the best question to ask?

Answer: A. "What happened to past students, with evidence?"

Ask for evidence.

- ✗ B. "What color is the certificate?": irrelevant
- ✗ C. "Is the teacher friendly?": not the key
- ✗ D. "Can I pay the full fee today to make sure?": no check

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_ftc_aicomply

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S106-F1-Q3 · question page 2219

AI.WORK.06 · Check job and course claims in practice

An app claims its "AI lawyer" can replace your attorney in court. What should you consider?

Answer: B. The claim may be untested or deceptive

Like claims the FTC challenged.

- ✗ A. That courts accept apps: not shown
- ✗ C. That lawyers are no longer needed: not shown
- ✗ D. That the claim must be true: not guaranteed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_ftc_aicomply

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S106-F1-Q4 · question page 2220

AI.WORK.06 · Check job and course claims in practice

Where should you look for reviews?

Answer: D. Independent sources, not only the seller's site

Independent reviews.

- ✗ A. Nowhere: reviews help
- ✗ B. Only on the seller's own website, since the seller knows the product better than anyone: biased
- ✗ C. Only in the ad: biased

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_ftc_aicomply

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S106-F1-Q5 · question page 2221

AI.WORK.06 · Check job and course claims in practice

Which offer seems most trustworthy?

Answer: C. A course with a clear syllabus, fair price and written refund policy

Clear and verifiable.

- ✗ A. An urgent offer ending today: red flag
- ✗ B. A course with no description, a large upfront fee and a promise of guaranteed wealth: red flags
- ✗ D. A secret system with testimonials only: red flags

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_ftc_aicomply

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S106-F2-Q1 · question page 2222

AI.WORK.06 · Research offers before paying

Which search helps check a company?

Answer: A. Its name plus "complaint" or "scam"

Look for problems.

- ✗ B. Its founder's birthday: irrelevant
- ✗ C. Its logo only: not useful
- ✗ D. Its name plus "best ever," which will show the most accurate reviews from customers: biased results

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_ftc_aicomply

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S106-F2-Q2 · question page 2223

AI.WORK.06 · Research offers before paying

What should you get in writing before paying?

Answer: C. The refund policy

Clear terms.

- ✗ A. Nothing: get it in writing
- ✗ B. The seller's favorite color: irrelevant
- ✗ D. A promise that you will become rich within one month of completing the course: not realistic

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_ftc_aicomply

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S106-F2-Q3 · question page 2224

AI.WORK.06 · Research offers before paying

Why compare with free or official resources?

Answer: A. To see whether the paid offer adds real value

Value check.

- ✗ B. Because paid offers are illegal: no
- ✗ C. To find a secret: no
- ✗ D. Because free resources are always worse than paid ones: not always

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_ftc_aicomply

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S106-F2-Q4 · question page 2225

AI.WORK.06 · Research offers before paying

Does AI change the rule "too good to be true usually is"?

Answer: B. No

Basic rules still apply.

- ✗ A. Yes, for online stores: no
- ✗ C. Only in 2024: no
- ✗ D. Yes, because AI makes all impossible promises possible, so old rules no longer apply: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_ftc_aicomply

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S106-F2-Q5 · question page 2226

AI.WORK.06 · Research offers before paying

A seller refuses to say what past students achieved. What does this suggest?

Answer: D. Results may not support the claims

Missing evidence.

- ✗ A. That the course is free: no

✗ B. Nothing at all: it is a warning sign

✗ C. That past students achieved even more than the advertisements say, but it is a secret: unsupported

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_ftc_aicomply

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S107-M-Q1 · question page 2228

AI.WORK.07 · Follow approved tools, protect confidential information and review output at work

Why might an employer allow some AI tools and not others?

Answer: C. Because of how they handle data

Data handling.

✗ A. Because of random choice: no

✗ B. Because all tools are identical: they differ

✗ D. Because some tools have nicer colors: not the reason

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S107-M-Q2 · question page 2229

AI.WORK.07 · Follow approved tools, protect confidential information and review output at work

Which is confidential information?

Answer: D. Customer account details

Private records.

✗ A. The company's public address: public

✗ B. A published recipe: public

✗ C. A public news article: public

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S107-M-Q3 · question page 2230

AI.WORK.07 · Follow approved tools, protect confidential information and review output at work

Who is responsible for AI-drafted text you send at work?

Answer: A. You

You send it, you own it.

- ✗ B. The recipient: no
- ✗ C. Nobody: someone is
- ✗ D. The AI tool: not true

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S107-M-Q4 · question page 2231

AI.WORK.07 · Follow approved tools, protect confidential information and review output at work

What should you do when AI rules at work are unclear?

Answer: B. Ask your manager

Ask first.

- ✗ A. Avoid all work: no
- ✗ C. Guess: risky
- ✗ D. Do whatever is fastest: risky

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S107-M-Q5 · question page 2232

AI.WORK.07 · Follow approved tools, protect confidential information and review output at work

Before AI output reaches others, what should you check?

Answer: D. Facts, tone and fit

Review before sending.

- ✗ A. Only the font: not enough

✗ B. Nothing: must check

✗ C. Only that it is long: not a check

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S107-B1-Q1 · question page 2233

AI.WORK.07 · Compare AI rules at work with rules about company keys and badges

Why does a badge open only some doors?

Answer: B. The company decides what access is safe

Controlled access.

✗ A. Because of the weather: no

✗ C. Because badges are broken and should open every door in the building for every worker: no

✗ D. Because doors are random: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S107-B1-Q2 · question page 2234

AI.WORK.07 · Compare AI rules at work with rules about company keys and badges

Approved AI tools are like what?

Answer: C. Doors your badge opens

Permitted access.

✗ A. Doors that are locked for everyone, including the managers and the owners of the company: not the comparison

✗ B. A broken lock: no

✗ D. A lost badge: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S107-B1-Q3 · question page 2235

AI.WORK.07 · Compare AI rules at work with rules about company keys and badges

Pasting private data into an unapproved tool is like what?

Answer: A. Propping open a secure door

Bypassing security.

- ✗ B. Reporting a problem: opposite
- ✗ C. Locking a door: opposite
- ✗ D. Using your badge on the doors: that is fine

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S107-B1-Q4 · question page 2236

AI.WORK.07 · Compare AI rules at work with rules about company keys and badges

Who is responsible for how you use your badge?

Answer: B. You

Personal responsibility.

- ✗ A. The badge maker: no
- ✗ C. The door, since the door decides who can go through it and why they are going: no
- ✗ D. Nobody: you are

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S107-B1-Q5 · question page 2237

AI.WORK.07 · Compare AI rules at work with rules about company keys and badges

What is the lesson?

Answer: D. Follow access rules for AI as for doors

Same principles.

- ✗ A. AI tools are completely different from every other kind of access at a workplace: similar
- ✗ B. Ignore rules when convenient: no
- ✗ C. Lend your badge freely: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S107-F1-Q1 · question page 2238

AI.WORK.07 · Use AI responsibly at work in practice

Sara wants to paste a customer complaint, including the customer's address, into an unapproved chatbot. What should she do?

Answer: C. Use an approved tool or remove the personal details

Protect personal data.

- ✗ A. Post it on social media: worse
- ✗ B. Email it to friends: worse
- ✗ D. Paste it, since customer addresses are not really private: confidential

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S107-F1-Q2 · question page 2239

AI.WORK.07 · Use AI responsibly at work in practice

An AI draft to a client promises a discount the company does not offer. What should happen?

Answer: A. Remove it before sending

Review for policy fit.

- ✗ B. Add more discounts: worse
- ✗ C. Send it to everyone: worse
- ✗ D. Send it: misleading

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S107-F1-Q3 · question page 2240

AI.WORK.07 · Use AI responsibly at work in practice

A team rule says to note when reports used AI drafting. What should you do?

Answer: C. Follow the rule and add the note

Follow disclosure rules.

- ✗ A. Ask AI to hide it: dishonest
- ✗ B. Skip the note, because nobody will notice whether AI was used to draft a report: breaks the rule
- ✗ D. Delete the report: unnecessary

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S107-F1-Q4 · question page 2241

AI.WORK.07 · Use AI responsibly at work in practice

Which use is clearly appropriate?

Answer: A. Drafting a public event announcement in an approved tool, then reviewing it

Approved, public, reviewed.

- ✗ B. Sharing passwords with a tool: never
- ✗ C. Sending unreviewed AI replies to clients: review first
- ✗ D. Pasting next year's secret product plans into a free public chatbot to get feedback: confidential

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S107-F1-Q5 · question page 2242

AI.WORK.07 · Use AI responsibly at work in practice

Which information should never go into an unapproved tool?

Answer: B. Passwords

Secrets stay out.

- ✗ A. A general grammar question: harmless
- ✗ C. A public holiday schedule: public
- ✗ D. The text of a public press release that the company already posted on its website: public

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S107-F2-Q1 · question page 2243

AI.WORK.07 · Handle gray areas and mistakes at work

You are unsure whether a document is confidential. What is the best approach?

Answer: D. Ask before using it in a tool

When unsure, ask.

- ✗ A. Paste half of it: still risky
- ✗ B. Assume it is public, since most workplace documents are not confidential anyway: risky
- ✗ C. Ignore the question: risky

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S107-F2-Q2 · question page 2244

AI.WORK.07 · Handle gray areas and mistakes at work

You accidentally pasted private data into the wrong tool. What should you do?

Answer: C. Report it promptly

Prompt reporting helps.

✗ A. Paste more data: worse

✗ B. Delete your account and say nothing: hides it

✗ D. Hide it and hope nobody finds out: makes it worse

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S107-F2-Q3 · question page 2245

AI.WORK.07 · Handle gray areas and mistakes at work

Which option is more careful in a gray area?

Answer: A. Leaving out the uncertain information

Choose caution.

✗ B. Guessing quickly: risky

✗ C. Including everything, since more information always leads to a better answer from the tool: riskier

✗ D. Using any tool: risky

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S107-F2-Q4 · question page 2246

AI.WORK.07 · Handle gray areas and mistakes at work

An AI summary says a policy allows refunds after 90 days, but the policy says 30. What does this show?

Answer: D. Output can conflict with company policy

Check against policy.

✗ A. That the policy must be outdated, because AI summaries are usually more current than policies: check the policy

✗ B. That refunds are not allowed: not shown

✗ C. Nothing important: it matters

Arithmetic checked by code: not applicable. Code example run and its output checked: not

applicable. Sources: none (stable)

S107-F2-Q5 · question page 2247

AI.WORK.07 · Handle gray areas and mistakes at work

Why is hiding a data mistake worse than reporting it?

Answer: B. It prevents the problem from being handled

Handle problems early.

- ✗ A. Because reports are long: not the reason
- ✗ C. Because managers like secrets: no
- ✗ D. It is not worse, since hiding a small mistake always makes it disappear on its own: not true

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S108-M-Q1 · question page 2249

AI.COST.01 · Compare free, subscription and pay-per-use pricing for a pattern of use

What does a subscription charge?

Answer: A. A fixed amount each period

Fixed recurring price.

- ✗ B. Nothing at all: that is free
- ✗ C. Only for what you actually use each month, so that light users always pay less: that is pay-per-use
- ✗ D. A one-time price forever: not a subscription

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S108-M-Q2 · question page 2250

AI.COST.01 · Compare free, subscription and pay-per-use pricing for a pattern of use

What does pay-per-use charge for?

Answer: C. What you actually use

Metered pricing.

- ✗ A. Nothing: it charges
- ✗ B. Only the first day: no
- ✗ D. A fixed amount every month: that is subscription

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S108-M-Q3 · question page 2251

AI.COST.01 · Compare free, subscription and pay-per-use pricing for a pattern of use

In this hypothetical example, what do 300 pay-per-use requests in a month cost?

Prepared example (hypothetical prices for practice; not any real product's prices)

Plan	Price	Notes
Free	0 dollars	Up to 20 requests a day
Subscription	20 dollars a month	No daily limit
Pay-per-use	0.05 dollars per request	No monthly fee

Answer: D. 15 dollars

300 x 0.05 = 15.

- ✗ A. 3 dollars: arithmetic slip
- ✗ B. 150 dollars: arithmetic slip
- ✗ C. 20 dollars: calculate

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: none (stable)

S108-M-Q4 · question page 2252

AI.COST.01 · Compare free, subscription and pay-per-use pricing for a pattern of use

In the same example, at how many requests a month do pay-per-use and the subscription cost the same?

Prepared example (hypothetical prices for practice; not any real product's prices)

Plan	Price	Notes
Free	0 dollars	Up to 20 requests a day
Subscription	20 dollars a month	No daily limit
Pay-per-use	0.05 dollars per request	No monthly fee

Answer: B. 400

20 / 0.05 = 400.

- ✗ A. 100: arithmetic slip
- ✗ C. 4,000: arithmetic slip
- ✗ D. 20: not related

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: none (stable)

S108-M-Q5 · question page 2253

AI.COST.01 · Compare free, subscription and pay-per-use pricing for a pattern of use

Which pattern usually fits a free plan?

Answer: C. Light, occasional use

Light use.

- ✗ A. Use by a large company: needs more
- ✗ B. Use that must never hit limits: free plans limit
- ✗ D. Heavy, all-day use for work that needs hundreds of requests every single day of the week: exceeds limits

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S108-B1-Q1 · question page 2254

AI.COST.01 · Compare AI plans with ways to pay for a gym or a bus

Paying per bus ride is like which plan?

Answer: B. Pay-per-use

Pay each time.

- ✗ A. Free plan: it costs
- ✗ C. A loan: no
- ✗ D. Subscription: that is per use

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S108-B1-Q2 · question page 2255

AI.COST.01 · Compare AI plans with ways to pay for a gym or a bus

A monthly bus pass is like which plan?

Answer: A. Subscription

Fixed price, unlimited use.

- ✗ B. A refund: no
- ✗ C. Free plan: it costs
- ✗ D. Pay-per-use: fixed price

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S108-B1-Q3 · question page 2256

AI.COST.01 · Compare AI plans with ways to pay for a gym or a bus

A free walking trail with fixed hours is like which plan?

Answer: D. Free plan

No cost, with limits.

- ✗ A. Pay-per-use: no charge
- ✗ B. A gift card: no
- ✗ C. Subscription: it is free

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S108-B1-Q4 · question page 2257

AI.COST.01 · Compare AI plans with ways to pay for a gym or a bus

You ride the bus twice a month. Which is likely cheaper?

Answer: B. Paying per ride

Light use.

- ✗ A. Neither: one is cheaper
- ✗ C. Buying a bus: no
- ✗ D. The monthly pass: not for light use

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S108-B1-Q5 · question page 2258

AI.COST.01 · Compare AI plans with ways to pay for a gym or a bus

What decides the best choice?

Answer: A. How often you use it

Your pattern of use.

- ✗ B. The driver's name: no
- ✗ C. The weather only: no
- ✗ D. The color of the bus: irrelevant

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S108-F1-Q1 · question page 2259

AI.COST.01 · Choose a pricing plan for a pattern of use

Mia uses an AI tool about 10 times a month, and the free plan allows 20 a day. Which plan fits?

Answer: C. The free plan

Well within free limits.

- ✗ A. Two subscriptions: unneeded
- ✗ B. An annual business plan: unneeded
- ✗ D. The subscription: unneeded

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S108-F1-Q2 · question page 2260

AI.COST.01 · Choose a pricing plan for a pattern of use

Hypothetical: a subscription costs 15 dollars a month; pay-per-use costs 0.10 dollars a request. Leo makes 300 requests a month. Which is cheaper?

Answer: D. The subscription

300 x 0.10 = 30 dollars, more than 15.

- ✗ A. They cost the same: 30 vs 15
- ✗ B. Neither can be compared: they can
- ✗ C. Pay-per-use: not here

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: none (stable)

S108-F1-Q3 · question page 2261

AI.COST.01 · Choose a pricing plan for a pattern of use

Same prices, but Ana makes 50 requests a month. Which is cheaper?

Answer: A. Pay-per-use

50 x 0.10 = 5 dollars.

- ✗ B. They cost the same: 5 vs 15
- ✗ C. Neither: one is cheaper
- ✗ D. The subscription: costs more here

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: none (stable)

S108-F1-Q4 · question page 2262

AI.COST.01 · Choose a pricing plan for a pattern of use

Which information do you need first to compare plans?

Answer: D. Your expected use per month

Pattern of use.

- ✗ A. The app's color: irrelevant
- ✗ B. The company's logo and the number of social media followers it has around the world: irrelevant
- ✗ C. The founder's name: irrelevant

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S108-F1-Q5 · question page 2263

AI.COST.01 · Choose a pricing plan for a pattern of use

A student hits the free plan's daily limit only during exam weeks. What is a reasonable choice?

Answer: C. Stay on the free plan or pay only during busy weeks

Match cost to use.

- ✗ A. Buy a yearly subscription right away, since hitting a limit once means you need it all year: overkill
- ✗ B. Make several free accounts to avoid limits: may break terms
- ✗ D. Stop studying: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S108-F2-Q1 · question page 2264

AI.COST.01 · Catch hidden costs and terms

Hypothetical: an annual plan costs 120 dollars; monthly costs 15 dollars. You will only use it for 2 months. Which is cheaper?

Answer: B. Monthly

2 x 15 = 30, less than 120.

- ✗ A. Neither: monthly is cheaper
- ✗ C. They cost the same: 120 vs 30
- ✗ D. Annual, since annual plans are always the cheaper option no matter how long you use the tool: not here

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: none (stable)

S108-F2-Q2 · question page 2265

AI.COST.01 · Catch hidden costs and terms

Why set a spending limit on pay-per-use?

Answer: C. Usage can spike and surprise you

Avoid surprise bills.

- ✗ A. Because limits are required by law: not the reason
- ✗ B. Because pay-per-use plans always charge the same amount every month no matter what: they vary
- ✗ D. To make the tool faster: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S108-F2-Q3 · question page 2266

AI.COST.01 · Catch hidden costs and terms

Do free and paid plans always have the same privacy terms?

Answer: D. Not always; check each

Terms can differ.

✗ A. Free plans have no terms: they do

✗ B. Paid plans have no terms: they do

✗ C. Yes: not always

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S108-F2-Q4 · question page 2267

AI.COST.01 · Catch hidden costs and terms

What should you check before subscribing?

Answer: B. Renewal and cancellation terms

Avoid surprises.

✗ A. Nothing: check terms

✗ C. Only the logo: irrelevant

✗ D. Only the price for the first month: may change

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S108-F2-Q5 · question page 2268

AI.COST.01 · Catch hidden costs and terms

A subscription includes ten features; you need one. What should you consider?

Answer: A. Whether a cheaper option covers that one feature

Pay for what you need.

✗ B. Nothing, since paying for unused features is always the best value for most people: not necessarily

✗ C. Using every feature anyway: not the point

✗ D. Buying a second subscription: worse

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S109-M-Q1 · question page 2270

AI.COST.02 · Estimate a metered cost from token counts and rates

What is a token?

Answer: B. A chunk of text a model reads or writes

Text pieces.

- ✗ A. A whole page: too large
- ✗ C. A password: no
- ✗ D. A coin that you must buy from the provider: not the meaning

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: tokens_openai

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S109-M-Q2 · question page 2271

AI.COST.02 · Estimate a metered cost from token counts and rates

By the rule of thumb, about how many tokens is 300 English words?

Answer: C. About 400

300 / 0.75 = 400.

- ✗ A. About 75: too few
- ✗ B. About 1,200: that is characters
- ✗ D. About 300: not exactly

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: tokens_openai

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S109-M-Q3 · question page 2272

AI.COST.02 · Estimate a metered cost from token counts and rates

In this hypothetical example, what do 500,000 input tokens cost?

Prepared example (hypothetical prices for practice; not any real product's prices)

Item	Hypothetical rate

Input tokens	2 dollars per million
Output tokens	8 dollars per million

Answer: A. 1 dollar

0.5 x 2 = 1.

- ✗ B. 4 dollars: that is output rate
- ✗ C. 10 dollars: arithmetic slip
- ✗ D. 2 dollars: half a million

Arithmetic checked by code: yes. Code example run and its output checked: not applicable.
Sources: tokens_openai

S109-M-Q4 · question page 2273

AI.COST.02 · Estimate a metered cost from token counts and rates

In the same example, what do 500,000 input plus 100,000 output tokens cost?

Prepared example (hypothetical prices for practice; not any real product's prices)

Item	Hypothetical rate
Input tokens	2 dollars per million
Output tokens	8 dollars per million

Answer: D. 1.80 dollars

1.00 + 0.80 = 1.80.

- ✗ A. 1.20 dollars: wrong output rate
- ✗ B. 5.00 dollars: arithmetic slip
- ✗ C. 1.00 dollar: output costs

Arithmetic checked by code: yes. Code example run and its output checked: not applicable.
Sources: tokens_openai

S109-M-Q5 · question page 2274

AI.COST.02 · Estimate a metered cost from token counts and rates

Which are often priced separately?

Answer: C. Input and output tokens

Different rates.

- ✗ A. Short and long words: no
- ✗ B. Morning and evening use: not usual
- ✗ D. Uppercase and lowercase letters, which always cost different amounts to process: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: tokens_openai

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S109-B1-Q1 · question page 2275

AI.COST.02 · Compute a per-million cost in three steps

2,000,000 tokens is how many millions?

Answer: D. 2

2,000,000 / 1,000,000 = 2.

- ✗ A. 200: no
- ✗ B. 0.2: no
- ✗ C. 20, because you count the number of zeros in the number and divide that by three: no

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: tokens_openai

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S109-B1-Q2 · question page 2276

AI.COST.02 · Compute a per-million cost in three steps

250,000 tokens is how many millions?

Answer: B. 0.25

250,000 / 1,000,000 = 0.25.

✗ A. 0.025: no

✗ C. 25, since there are two hundred and fifty thousand tokens, which rounds to twenty-five: no

✗ D. 2.5: no

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: tokens_openai

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S109-B1-Q3 · question page 2277

AI.COST.02 · Compute a per-million cost in three steps

Input cost at 3 dollars per million for 2 million?

Answer: A. 6 dollars

2 x 3 = 6.

✗ B. 5 dollars: added

✗ C. 1.50 dollars: divided

✗ D. 3 dollars: multiply

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: tokens_openai

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S109-B1-Q4 · question page 2278

AI.COST.02 · Compute a per-million cost in three steps

Output cost at 12 per million for 0.25 million?

Answer: C. 3 dollars

0.25 x 12 = 3.

✗ A. 48 dollars: multiplied wrong

✗ B. 0.30 dollars: no

✗ D. 12 dollars: multiply

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: tokens_openai

S109-B1-Q5 · question page 2279

AI.COST.02 · Compute a per-million cost in three steps

What is the total?

Answer: B. 9 dollars

6 + 3 = 9.

- ✗ A. 18 dollars: no
- ✗ C. 15 dollars: added rates
- ✗ D. 6 dollars: add output

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: tokens_openai

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S109-F1-Q1 · question page 2280

AI.COST.02 · Estimate token costs in practice

Hypothetical rates of 2 dollars per million input and 8 per million output. A job uses 3,000,000 input and 500,000 output tokens. What is the cost?

Answer: D. 10 dollars

6 + 4 = 10.

- ✗ A. 11 dollars: arithmetic slip
- ✗ B. 20 dollars: arithmetic slip
- ✗ C. 6 dollars: add output

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: tokens_openai

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S109-F1-Q2 · question page 2281

AI.COST.02 · Estimate token costs in practice

A 1,000-word document is sent as input. About how many tokens is that, by the rule of thumb?

Answer: A. About 1,333

1,000 / 0.75 ≈ 1,333.

- ✗ B. About 250: too few
- ✗ C. About 4,000: that is characters
- ✗ D. Exactly 1,000: not exact

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: tokens_openai

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S109-F1-Q3 · question page 2282

AI.COST.02 · Estimate token costs in practice

Which change most reduces output cost?

Answer: D. Asking for shorter answers

Fewer output tokens.

- ✗ A. Adding more attachments: raises input
- ✗ B. Repeating the question: raises input
- ✗ C. Typing your question: no effect

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: tokens_openai

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S109-F1-Q4 · question page 2283

AI.COST.02 · Estimate token costs in practice

A tool says your request used 12,000 input tokens. Is that the same as 12,000 words?

Answer: A. No; tokens are not words

Rule of thumb only.

- ✗ B. Only for short texts: no
- ✗ C. Yes, in English only: still differ
- ✗ D. Yes: they differ

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: tokens_openai

S109-F1-Q5 · question page 2284

AI.COST.02 · Estimate token costs in practice

Hypothetical: 100 requests, each 2,000 input and 500 output tokens, at 2 and 8 dollars per million. Total cost?

Answer: C. 0.80 dollars

200,000 input = 0.40; 50,000 output = 0.40.

- ✗ A. 0.40 dollars: missed output
- ✗ B. 80 dollars: arithmetic slip
- ✗ D. 8.00 dollars: arithmetic slip

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: tokens_openai

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S109-F2-Q1 · question page 2285

AI.COST.02 · Avoid common estimation mistakes

Why do long conversations cost more per message over time?

Answer: B. Earlier messages are resent as input

Input grows each turn.

- ✗ A. Because tokens get bigger: no
- ✗ C. Because answers get shorter: not the reason
- ✗ D. Because providers charge a fee: not the reason

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: tokens_openai

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S109-F2-Q2 · question page 2286

AI.COST.02 · Avoid common estimation mistakes

Do attached documents count toward input tokens?

Answer: A. Yes

Attachments are input.

- ✗ B. No, because attachments are always processed for free by every provider of every tool: they count
- ✗ C. Only on weekends: no
- ✗ D. Only images: documents count too

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: tokens_openai

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S109-F2-Q3 · question page 2287

AI.COST.02 · Avoid common estimation mistakes

Output costs 4 times input per token. Which request is likely most expensive?

Answer: C. A short question asking for a very long report

Long output at a higher rate.

- ✗ A. No request: free
- ✗ B. A short question with a short answer: cheap
- ✗ D. A long question asking for a one-word answer: output dominates

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: tokens_openai

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S109-F2-Q4 · question page 2288

AI.COST.02 · Avoid common estimation mistakes

Before a big job, what is the best way to estimate cost?

Answer: B. Test a small sample and measure tokens

Measure, then multiply.

- ✗ A. Guess the total from the length of the instructions without running any test at all: unreliable
- ✗ C. Ask a friend: unreliable
- ✗ D. Assume it is free: risky

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: tokens_openai

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S109-F2-Q5 · question page 2289

AI.COST.02 · Avoid common estimation mistakes

Hypothetical: a 10-request sample used 50,000 tokens. About how many tokens will 1,000 similar requests use?

Answer: D. 5,000,000

50,000 x 100 = 5,000,000.

- ✗ A. 50,000,000: arithmetic slip
- ✗ B. 500,000: arithmetic slip
- ✗ C. 50,000: scale up

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: tokens_openai

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S110-M-Q1 · question page 2291

AI.COST.03 · Plan around usage limits and response time

What is a usage limit?

Answer: A. A cap on how much you can use a tool

Caps on use.

- ✗ B. The tool's price: related but different
- ✗ C. The tool's name: no
- ✗ D. A rule that says how fast you must type when you send a message to the tool: not the meaning

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S110-M-Q2 · question page 2292

AI.COST.03 · Plan around usage limits and response time

What is latency?

Answer: D. Waiting time before the answer arrives

Response delay.

- ✗ A. The tool's cost: no
- ✗ B. The answer's length: no
- ✗ C. How accurate the answer is compared with the correct answer from a trusted source: not latency

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S110-M-Q3 · question page 2293

AI.COST.03 · Plan around usage limits and response time

What usually increases latency?

Answer: B. Larger models and longer answers

Bigger and longer means slower.

- ✗ A. Smaller models: faster
- ✗ C. Shorter questions sent during quiet times of day when few other people are using the tool: reduces it
- ✗ D. Short answers: faster

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S110-M-Q4 · question page 2294

AI.COST.03 · Plan around usage limits and response time

When should you do heavy AI work before a deadline?

Answer: C. Early

Leave time for limits.

✗ A. After the deadline: too late

✗ B. Right before the deadline, since tools are always fastest when many people are using them: risky

✗ D. Never: not necessary

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S110-M-Q5 · question page 2295

AI.COST.03 · Plan around usage limits and response time

A live chat needs quick replies. What should you choose?

Answer: D. A faster option, even if slightly less capable

Match speed to need.

✗ A. No tool: not necessary

✗ B. A tool with long waits: too slow

✗ C. The slowest and largest model available: too slow

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S110-B1-Q1 · question page 2296

AI.COST.03 · Compare usage limits with a library's borrowing rules

A 5-book borrowing cap is like what?

Answer: B. A usage limit

A cap on use.

✗ A. A token: no

✗ C. A subscription: no

✗ D. Latency: that is a cap

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S110-B1-Q2 · question page 2297

AI.COST.03 · Compare usage limits with a library's borrowing rules

Waiting for a popular book is like what?

Answer: A. Latency at busy times

Delay.

- ✗ B. A usage limit, because the library refuses to lend popular books to anyone at all: that is waiting
- ✗ C. A refund: no
- ✗ D. A free plan: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S110-B1-Q3 · question page 2298

AI.COST.03 · Compare usage limits with a library's borrowing rules

Why go to the library early in the week?

Answer: C. To avoid being stuck near the deadline

Plan ahead.

- ✗ A. Because books are cheaper on Monday: no
- ✗ B. To meet friends: not the point
- ✗ D. Because libraries close every Thursday night: not the reason

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S110-B1-Q4 · question page 2299

AI.COST.03 · Compare usage limits with a library's borrowing rules

What is like a backup plan?

Answer: A. Knowing another library or source

Alternatives.

- ✗ B. Borrowing more than allowed: breaks rules
- ✗ C. Giving up: no
- ✗ D. Waiting at the same library: no backup

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S110-B1-Q5 · question page 2300

AI.COST.03 · Compare usage limits with a library's borrowing rules

What is the lesson for AI tools?

Answer: C. Plan ahead for limits and waits

Planning.

- ✗ A. Never use tools: too strong
- ✗ B. Only use them at night: not the lesson
- ✗ D. Assume tools are always instantly available with no limits, especially near deadlines: risky

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S110-F1-Q1 · question page 2301

AI.COST.03 · Work around limits and latency in practice

Sam's free plan allows 30 messages a day. His project needs about 90. What is a sensible plan?

Answer: D. Spread the work over at least 3 days

90 / 30 = 3 days.

- ✗ A. Give up: unnecessary
- ✗ B. Create many fake accounts: may break terms
- ✗ C. Do it all on the last day, since limits are rarely enforced when many messages are needed: risky

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: none (stable)

S110-F1-Q2 · question page 2302

AI.COST.03 · Work around limits and latency in practice

A tool slows down every evening when many people use it. What helps?

Answer: B. Doing heavy work at quieter times

Avoid peak times.

- ✗ A. Using longer prompts: slower
- ✗ C. Asking for longer answers: slower
- ✗ D. Sending the same request many times: worse

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S110-F1-Q3 · question page 2303

AI.COST.03 · Work around limits and latency in practice

An upload fails because the file is over the size limit. What should you do?

Answer: C. Split or shrink the file

Respect the limit.

- ✗ A. Give up on the task: unnecessary
- ✗ B. Rename it: not a fix
- ✗ D. Upload the same file again: will not work

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S110-F1-Q4 · question page 2304

AI.COST.03 · Work around limits and latency in practice

A store chatbot takes 40 seconds to reply. What problem does this cause?

Answer: D. Customers may leave before getting help

Latency matters for live use.

- ✗ A. The store saves money: not the point
- ✗ B. None, since customers always wait patiently for as long as a chatbot needs to answer: not realistic
- ✗ C. The answers become more accurate: not caused

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S110-F1-Q5 · question page 2305

AI.COST.03 · Work around limits and latency in practice

A report is due tomorrow morning and needs AI help. When should you start?

Answer: B. Today, leaving time for limits or delays

Buffer time.

- ✗ A. Never: unnecessary
- ✗ C. After the deadline: too late
- ✗ D. Tomorrow morning, a few minutes before the deadline, since tools respond instantly: risky

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S110-F2-Q1 · question page 2306

AI.COST.03 · Build resilience against changing limits and outages

A free plan's limit was lowered without much notice. What does this show?

Answer: A. Limits can change

Plan for change.

- ✗ B. That you did something wrong: not necessarily
- ✗ C. That the tool is broken: not necessarily

✗ D. That limits are fixed by law: they can change

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S110-F2-Q2 · question page 2307

AI.COST.03 · Build resilience against changing limits and outages

An automation calls a tool 500 times a minute and starts failing. What is the likely cause?

Answer: D. A rate limit

Too many requests per minute.

✗ A. The tool is free: irrelevant

✗ B. A spelling mistake in the automation's name that makes the tool reject every request: not likely

✗ C. The answers are too short: irrelevant

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S110-F2-Q3 · question page 2308

AI.COST.03 · Build resilience against changing limits and outages

How should an automation respond to a rate limit?

Answer: B. Wait and retry slowly

Back off and retry.

✗ A. Stop forever: too drastic

✗ C. Delete its logs: unhelpful

✗ D. Send even more requests immediately, so that the tool is forced to answer them faster: worse

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S110-F2-Q4 · question page 2309

AI.COST.03 · Build resilience against changing limits and outages

Why save your work as you go?

Answer: C. So a limit or outage does not erase progress

Protect progress.

- ✗ A. Because tools require it: not the reason
- ✗ B. To use more storage: no
- ✗ D. Because saving makes the AI tool answer faster: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S110-F2-Q5 · question page 2310

AI.COST.03 · Build resilience against changing limits and outages

For an important deadline, what is a wise approach?

Answer: A. Have a backup in case the tool is unavailable

Resilience.

- ✗ B. Wait until the last minute: risky
- ✗ C. Depend completely on one tool, since popular tools are always available at every moment: risky
- ✗ D. Ignore the deadline: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S111-M-Q1 · question page 2312

AI.COST.04 · Compare running a model through a provider with running one locally

What happens when you use a model through a provider?

Answer: B. Your request goes to the provider's computers

Cloud processing.

✗ A. Nothing is sent anywhere: data is sent

✗ C. You must build the model: no

✗ D. The model always runs entirely on your own phone: that is local

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S111-M-Q2 · question page 2313

AI.COST.04 · Compare running a model through a provider with running one locally

What is a benefit of running a model locally?

Answer: D. Data can stay on your device

Privacy and offline use.

✗ A. Updates happen automatically: your job

✗ B. You always get the most powerful models available, which are larger than any cloud model: usually smaller

✗ C. No hardware is needed: hardware needed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S111-M-Q3 · question page 2314

AI.COST.04 · Compare running a model through a provider with running one locally

What is a cost of running locally?

Answer: A. Capable hardware and your own upkeep

Hardware and maintenance.

✗ B. Data always leaves your device: not necessarily

✗ C. It cannot work offline: it can

✗ D. A monthly fee paid to the provider: not local

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S111-M-Q4 · question page 2315

AI.COST.04 · Compare running a model through a provider with running one locally

What is a benefit of using a provider?

Answer: C. Powerful models without special hardware

Capability and convenience.

- ✗ A. It is always free: not always
- ✗ B. It works with no internet: needs internet
- ✗ D. Your data never leaves your device: it does leave

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S111-M-Q5 · question page 2316

AI.COST.04 · Compare running a model through a provider with running one locally

Who handles updates for a local model?

Answer: B. You

Your responsibility.

- ✗ A. The internet: no
- ✗ C. The provider, who updates every downloaded model automatically from its own computers: not usually
- ✗ D. Nobody; models never need updates: they do

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S111-B1-Q1 · question page 2317

AI.COST.04 · Compare cloud and local models with eating out and cooking at home

A restaurant is like what?

Answer: D. Using a provider

Someone else's kitchen.

- ✗ A. A token: no
- ✗ B. A subscription only: not the comparison
- ✗ C. Running a model locally: that is home

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S111-B1-Q2 · question page 2318

AI.COST.04 · Compare cloud and local models with eating out and cooking at home

Cooking at home is like what?

Answer: C. Running a model locally

Your own kitchen.

- ✗ A. A rate limit: no
- ✗ B. Using a provider, since your kitchen is connected to the restaurant through the internet: no
- ✗ D. A free trial: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S111-B1-Q3 · question page 2319

AI.COST.04 · Compare cloud and local models with eating out and cooking at home

What do you need to cook at home?

Answer: A. A kitchen and skills

Like hardware and setup.

- ✗ B. Nothing at all, since home cooking needs no equipment or knowledge of any kind: no
- ✗ C. A waiter: no
- ✗ D. A restaurant: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S111-B1-Q4 · question page 2320

AI.COST.04 · Compare cloud and local models with eating out and cooking at home

What does home cooking give you?

Answer: B. Control over ingredients

Like data control.

- ✗ A. Access to a team of professional chefs who cook every meal for you every day: that is a restaurant
- ✗ C. No dishes to wash: you clean up
- ✗ D. Free meals forever: still costs

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S111-B1-Q5 · question page 2321

AI.COST.04 · Compare cloud and local models with eating out and cooking at home

Which is always better?

Answer: C. Neither; it depends on the meal

Depends on needs.

- ✗ A. Fast food only: no
- ✗ B. Home, always: not always
- ✗ D. Restaurants, because professional kitchens always make every meal better than home kitchens: not always

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S111-F1-Q1 · question page 2322

AI.COST.04 · Choose cloud or local in practice

A clinic wants to summarize notes containing patient details and has strict rules about data leaving its building. Which option deserves serious consideration?

Answer: A. A local model, if it is capable enough

Data stays on site.

- ✗ B. Posting notes online: no
- ✗ C. Emailing notes to a stranger: no
- ✗ D. Any free public chatbot, since summarizing notes is a simple task with no privacy concerns: data leaves

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S111-F1-Q2 · question page 2323

AI.COST.04 · Choose cloud or local in practice

A student with an old laptop wants the most capable model for essays. Which fits?

Answer: D. A provider

No special hardware needed.

- ✗ A. Building a data center: no
- ✗ B. No tool: not necessary
- ✗ C. A large local model: hardware limits

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S111-F1-Q3 · question page 2324

AI.COST.04 · Choose cloud or local in practice

A field researcher works where there is no internet. Which option can work?

Answer: C. A local model

Offline use.

✗ A. Only a phone call: no

✗ B. Neither: local can

✗ D. A cloud provider: needs internet

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S111-F1-Q4 · question page 2325

AI.COST.04 · Choose cloud or local in practice

A small team has no one to maintain software. Which is easier for them?

Answer: A. A provider

Provider handles upkeep.

✗ B. Building their own model: harder

✗ C. Neither: one is easier

✗ D. Running local models: they do

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S111-F1-Q5 · question page 2326

AI.COST.04 · Choose cloud or local in practice

Which question matters most when choosing?

Answer: B. "How sensitive is the data?"

Sensitivity drives the choice.

✗ A. "Which is newest?": not the key

✗ C. "Which do my friends use?": not the key

✗ D. "Which option has the more attractive logo and the more popular brand name online?": irrelevant

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S111-F2-Q1 · question page 2327

AI.COST.04 · Avoid assumptions about local and cloud

Is a local app automatically private?

Answer: D. No; some still send data online

Check its settings.

- ✗ A. Yes, if it is free: no
- ✗ B. Only on phones: no
- ✗ C. Yes, because every app that runs on your own device keeps all data on the device forever: not always

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S111-F2-Q2 · question page 2328

AI.COST.04 · Avoid assumptions about local and cloud

Is running a large model locally automatically free?

Answer: A. No; hardware and electricity cost money

Hidden costs.

- ✗ B. Yes, on weekends: no
- ✗ C. Only for students: no
- ✗ D. Yes, because once a model is downloaded it can be run forever without any cost at all: costs remain

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S111-F2-Q3 · question page 2329

AI.COST.04 · Avoid assumptions about local and cloud

Who must install security fixes for a local setup?

Answer: D. You

Your responsibility.

- ✗ A. Nobody: needed
- ✗ B. Your internet company: no
- ✗ C. The provider of a cloud service, who reaches into your device to fix it automatically: not local

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S111-F2-Q4 · question page 2330

AI.COST.04 · Avoid assumptions about local and cloud

Is cloud use automatically unsafe?

Answer: B. No; providers may offer strong security and controls

Evaluate the provider.

- ✗ A. Only at night: no
- ✗ C. Yes, always: too strong
- ✗ D. Yes, because any data that leaves your device is always stolen: too strong

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S111-F2-Q5 · question page 2331

AI.COST.04 · Avoid assumptions about local and cloud

Which factors should guide the choice?

Answer: C. Data sensitivity, capability, budget and maintenance

Several factors.

- ✗ A. Only price: too narrow
- ✗ B. Only speed: too narrow

✗ D. Only which option sounds more modern and impressive: irrelevant

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S112-M-Q1 · question page 2333

AI.COST.05 · Explain that downloadable weights still come with license terms

What are a model's weights?

Answer: B. The learned numbers that make it work

Learned numbers.

✗ A. The training data: related but different

✗ C. The model's price: no

✗ D. How heavy the computer is that runs the model, measured in kilograms or pounds: not the meaning

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: tech_osaid

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S112-M-Q2 · question page 2334

AI.COST.05 · Explain that downloadable weights still come with license terms

What does "open weights" mean?

Answer: D. Anyone can download the weights

Downloadable.

✗ A. Anyone can use the model for any purpose at all, with no license or rules of any kind: license applies

✗ B. The weights are secret: opposite

✗ C. The model is broken: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: tech_osaid

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S112-M-Q3 · question page 2335

AI.COST.05 · Explain that downloadable weights still come with license terms

In this fictional license, may you use the model in a paid product without another agreement?

Prepared example (a fictional license excerpt)

Model license (excerpt): "You may use, copy and modify this model for research and personal projects. Commercial use requires a separate written agreement. You must keep this notice in any copy you share."

Answer: C. No

Commercial use needs a separate agreement.

- ✗ A. Yes, if you modify it: still commercial
- ✗ B. Yes, after one month: no
- ✗ D. Yes: license limits it

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: tech_osaid

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S112-M-Q4 · question page 2336

AI.COST.05 · Explain that downloadable weights still come with license terms

In the same license, what must you do when sharing a copy?

Prepared example (a fictional license excerpt)

Model license (excerpt): "You may use, copy and modify this model for research and personal projects. Commercial use requires a separate written agreement. You must keep this notice in any copy you share."

Answer: A. Keep the notice

Stated requirement.

- ✗ B. Pay a fee: not stated
- ✗ C. Nothing: keep the notice

✗ D. Remove the notice: violates terms

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: tech_osaid

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S112-M-Q5 · question page 2337

AI.COST.05 · Explain that downloadable weights still come with license terms

Are open weights automatically "open source"?

Answer: D. No, not automatically

Definitions differ and are debated.

✗ A. Yes, if they are free: no
✗ B. Only for small models: no
✗ C. Yes: not automatically

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: tech_osaid

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S112-B1-Q1 · question page 2338

AI.COST.05 · Compare model licenses with a free font's terms

Is the font free to download?

Answer: A. Yes

Downloading is free.

✗ B. Only for companies: no
✗ C. Only on weekends: no
✗ D. No, because every font on the internet must be bought before it can be downloaded: it is free

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: tech_osaid

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S112-B1-Q2 · question page 2339

AI.COST.05 · Compare model licenses with a free font's terms

May the designer use it in paid ads without a license?

Answer: B. No

Commercial use needs a license.

- ✗ A. Yes, if the ad is small: still commercial
- ✗ C. Yes, after a week: no
- ✗ D. Yes: terms limit it

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: tech_osaid

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S112-B1-Q3 · question page 2340

AI.COST.05 · Compare model licenses with a free font's terms

What decides what the designer may do?

Answer: C. The terms

License terms.

- ✗ A. Where it was found: not by itself
- ✗ B. The font's color: irrelevant
- ✗ D. The size of the font file, since smaller fonts always come with fewer rules: irrelevant

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: tech_osaid

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S112-B1-Q4 · question page 2341

AI.COST.05 · Compare model licenses with a free font's terms

Open-weight models are like the font how?

Answer: D. Free to download, but use depends on the license

Same principle.

- ✗ A. They have no files: no

✗ B. They are completely different, because model licenses are never enforced by anyone: not true

✗ C. They are always paid: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: tech_osaid

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S112-B1-Q5 · question page 2342

AI.COST.05 · Compare model licenses with a free font's terms

What should you do before using either one in a project?

Answer: C. Read the terms for your intended use

Read the license.

✗ A. Ask a friend whether they think the terms probably allow it, and go with their guess: unreliable

✗ B. Assume it is allowed: risky

✗ D. Delete the terms: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: tech_osaid

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S112-F1-Q1 · question page 2343

AI.COST.05 · Read license terms for an intended use

A student uses an open-weight model for a personal study app. The license allows personal use. Is this allowed?

Answer: B. Yes, under that license

Matches the terms.

✗ A. No, because open-weight models can never be used by students for any purpose at all: license allows it

✗ C. Only with a lawyer: not required here

✗ D. Only if paid: not required

Arithmetic checked by code: not applicable. Code example run and its output checked: not

applicable. Sources: tech_osaid

S112-F1-Q2 · question page 2344

AI.COST.05 · Read license terms for an intended use

A startup wants to sell an app built on a model whose license says "non-commercial use only." What follows?

Answer: A. They need different permission or a different model

Respect the license.

- ✗ B. They can sell it if they change the app's name: still commercial
- ✗ C. They can ignore it: no
- ✗ D. Nothing: not stated

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: tech_osaid

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S112-F1-Q3 · question page 2345

AI.COST.05 · Read license terms for an intended use

A license requires credit to the model's creators. Where should credit appear?

Answer: D. Where the license says

Follow the terms.

- ✗ A. In a secret file: not visible
- ✗ B. Only in your diary: no
- ✗ C. Nowhere: required

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: tech_osaid

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S112-F1-Q4 · question page 2346

AI.COST.05 · Read license terms for an intended use

Two versions of a model have different licenses. Which applies?

Answer: A. The license for the version you use

Version-specific.

- ✗ B. The oldest license, because the first version's license always applies to every later one: not necessarily
- ✗ C. The shortest: no
- ✗ D. Neither: one applies

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: tech_osaid

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S112-F1-Q5 · question page 2347

AI.COST.05 · Read license terms for an intended use

Someone says, "It's open weights, so there are no rules." What is more accurate?

Answer: C. Its license sets the rules

License terms apply.

- ✗ A. Only governments set rules: licenses too
- ✗ B. Only the app store sets rules: not the main point
- ✗ D. That is right; open-weight models are always released with no license or conditions: licenses exist

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: tech_osaid

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S112-F2-Q1 · question page 2348

AI.COST.05 · Handle varied and changing license terms

A license forbids use in products with more than a certain number of users. Why does this matter?

Answer: B. A growing product might later break the terms

Plan for growth.

- ✗ A. It removes the license: no

✗ C. It makes the model faster: no

✗ D. It does not matter, because user limits in licenses are never enforced against anyone: it matters

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: tech_osaid

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S112-F2-Q2 · question page 2349

AI.COST.05 · Handle varied and changing license terms

A license requires sharing your changes to the model. What should you do if you modify it?

Answer: D. Share as the license requires

Follow conditions.

✗ A. Rename the model: still applies

✗ B. Delete the license: no

✗ C. Keep the changes secret: not stated

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: tech_osaid

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S112-F2-Q3 · question page 2350

AI.COST.05 · Handle varied and changing license terms

The terms for a public project are unclear. What is the best step?

Answer: A. Ask someone who handles legal questions

Get advice.

✗ B. Ignore the terms: risky

✗ C. Copy another project: not permission

✗ D. Guess the most generous meaning of the terms: risky

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: tech_osaid

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S112-F2-Q4 · question page 2351

AI.COST.05 · Handle varied and changing license terms

Is "everyone else uses it" permission to use a model however you like?

Answer: B. No

Popularity is not permission.

- ✗ A. Only online: no
- ✗ C. Yes, after a year: no
- ✗ D. Yes: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: tech_osaid

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S112-F2-Q5 · question page 2352

AI.COST.05 · Handle varied and changing license terms

Is the Open Source AI Definition accepted by everyone?

Answer: C. No; some groups dispute it

An ongoing debate.

- ✗ A. It does not exist: it exists
- ✗ B. Yes, every company and every group in the world has fully agreed to it without any debate: disputed
- ✗ D. It is a law: it is a definition

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: tech_osaid

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S113-M-Q1 · question page 2354

AI.COST.06 · Choose a tool from stated needs, limits and costs

Where should tool choice start?

Answer: B. With the task

Task first.

✗ A. With whichever brand is most famous, since famous tools are always best for every task: not the method

✗ C. With a random pick: no

✗ D. With the newest tool: not the method

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S113-M-Q2 · question page 2355

AI.COST.06 · Choose a tool from stated needs, limits and costs

In this hypothetical example, a student needs to read PDFs with a school-approved tool. Which fits?

Prepared example (hypothetical prices for practice; not any real product's prices)

Need or limit	Tool A	Tool B	Tool C
Reads PDFs	Yes	Yes	No
Approved by school	No	Yes	Yes
Monthly cost (hypothetical)	10 dollars	0 dollars	0 dollars

Answer: A. Tool B

Meets both needs, at no cost.

✗ B. Tool A, because it reads PDFs and the school's approval does not matter for homework: not approved

✗ C. None of them: B fits

✗ D. Tool C: no PDFs

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S113-M-Q3 · question page 2356

AI.COST.06 · Choose a tool from stated needs, limits and costs

What are switching costs?

Answer: C. Time and effort to move to a new tool

Learning and migration.

- ✗ A. A cancellation fee only: too narrow
- ✗ B. The cost of electricity: no
- ✗ D. The monthly price of the new tool: more than price

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S113-M-Q4 · question page 2357

AI.COST.06 · Choose a tool from stated needs, limits and costs

How should you judge quality?

Answer: D. Try it on a few real examples

Real tasks.

- ✗ A. Look at its logo: irrelevant
- ✗ B. Check how many downloads it has: popularity
- ✗ C. Read the tool's own advertisements: biased

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S113-M-Q5 · question page 2358

AI.COST.06 · Choose a tool from stated needs, limits and costs

Which is a common waste?

Answer: A. Paying for several similar tools

Overlapping subscriptions.

- ✗ B. Reading the terms: good practice
- ✗ C. Testing tools on real tasks: good practice
- ✗ D. Using a free tool: not waste

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S113-B1-Q1 · question page 2359

AI.COST.06 · Compare options with a needs table

Which option fails on cost?

Answer: C. X

8 is over 5.

- ✗ A. Y: it is free
- ✗ B. Z, because three dollars a month is more than the budget of five dollars a month: 3 is under 5
- ✗ D. None: X fails

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: none (stable)

S113-B1-Q2 · question page 2360

AI.COST.06 · Compare options with a needs table

Which option fails "works offline"?

Answer: D. Y

Y is not offline.

- ✗ A. None: Y fails
- ✗ B. Z: works offline
- ✗ C. X, since paid options never work offline because they need to check your payment: X works offline

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S113-B1-Q3 · question page 2361

AI.COST.06 · Compare options with a needs table

Which option should be chosen?

Answer: B. Z

Passes both must-haves.

- ✗ A. Y, because free options are always the best choice even when they fail a must-have need: fails offline
- ✗ C. None: Z passes
- ✗ D. X: over budget

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S113-B1-Q4 · question page 2362

AI.COST.06 · Compare options with a needs table

What do you do with an option that fails a must-have?

Answer: A. Cross it out

Must-haves are firm.

- ✗ B. Ignore the need: no
- ✗ C. Pay extra: no
- ✗ D. Choose it anyway: fails the need

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S113-B1-Q5 · question page 2363

AI.COST.06 · Compare options with a needs table

Among options that pass, which should you prefer?

Answer: C. The cheapest that does the job well

Value.

- ✗ A. The one with the longest name: no

✗ B. The newest: not the rule

✗ D. The most expensive: not necessarily

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S113-F1-Q1 · question page 2364

AI.COST.06 · Choose tools sensibly in practice

Lia's school provides an approved AI tool that meets her needs. An ad pushes a 15-dollar app. What should she do?

Answer: B. Use the approved tool she already has

Avoid unneeded subscriptions.

✗ A. Use neither: unnecessary

✗ C. Buy the app, since paid apps are always better than tools that schools provide for free: unneeded

✗ D. Use both and pay for both: waste

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S113-F1-Q2 · question page 2365

AI.COST.06 · Choose tools sensibly in practice

Ben uses a tool daily, and his saved prompts and files are in it. Another tool is slightly better. What should he weigh?

Answer: D. Switching costs against the improvement

Effort to move.

✗ A. Nothing, because switching tools is always free and instant, with no time or effort at all: it costs

✗ B. Only the logo: irrelevant

✗ C. Only the price: too narrow

Arithmetic checked by code: not applicable. Code example run and its output checked: not

applicable. Sources: none (stable)

S113-F1-Q3 · question page 2366

AI.COST.06 · Choose tools sensibly in practice

Which is the best way to compare two writing tools?

Answer: B. Test both on the same real writing tasks

Real-task comparison.

- ✗ A. Choose the one with more downloads: popularity
- ✗ C. Choose the one with the newest version number: irrelevant
- ✗ D. Compare their advertisements: biased

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S113-F1-Q4 · question page 2367

AI.COST.06 · Choose tools sensibly in practice

A club pays for three AI tools that all summarize documents. What should it do?

Answer: C. Keep one that meets its needs and cancel the others

Remove overlap.

- ✗ A. Add a fourth tool so that every club member has a different summarizer to choose from: more waste
- ✗ B. Keep all three: waste
- ✗ D. Cancel all three without checking: may lose needed one

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S113-F1-Q5 · question page 2368

AI.COST.06 · Choose tools sensibly in practice

Which need is a "limit" rather than a feature?

Answer: D. A budget of 0 dollars

Constraint.

- ✗ A. Supporting Spanish: feature
- ✗ B. Making charts: feature
- ✗ C. Being able to read PDF files that are uploaded by users of the tool during the day: that is a feature

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S113-F2-Q1 · question page 2369

AI.COST.06 · Resist hype and avoid lock-in

A vendor's demo looks amazing. What should you do before buying?

Answer: A. Test it on your own real tasks

Your tasks, not theirs.

- ✗ B. Ask the vendor if it works: biased
- ✗ C. Watch the demo again: not a test
- ✗ D. Buy it immediately: demos are chosen

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S113-F2-Q2 · question page 2370

AI.COST.06 · Resist hype and avoid lock-in

What helps with free trials?

Answer: B. A reminder to cancel before it renews

Avoid surprise charges.

- ✗ A. Ignoring the renewal date: risky
- ✗ C. Signing up for as many free trials as possible at the same time to compare them all quickly: risky

✗ D. Sharing your card number: risky

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S113-F2-Q3 · question page 2371

AI.COST.06 · Resist hype and avoid lock-in

How often should you review subscriptions?

Answer: D. Every few months

Regular review.

✗ A. Only once in your life: too rare
✗ B. Every hour: too often
✗ C. Never, because subscriptions always cancel themselves when you stop using the tool: they renew

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S113-F2-Q4 · question page 2372

AI.COST.06 · Resist hype and avoid lock-in

A tool is central to your work. What should you check?

Answer: A. How to export your data if you leave

Avoid lock-in.

✗ B. Nothing: lock-in risk
✗ C. Its founder's name: irrelevant
✗ D. Whether its logo will look good: irrelevant

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S113-F2-Q5 · question page 2373

AI.COST.06 · Resist hype and avoid lock-in

What question cuts through marketing?

Answer: C. "What problem does it solve that my tools do not?"

Need-based question.

- ✗ A. "Is it expensive?": not the key
- ✗ B. "Does it have a nice website?": irrelevant
- ✗ D. "Is it the newest and most talked-about AI tool?": hype

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S114-M-Q1 · question page 2375

AI.EVAL.01 · Build a small test set with expected answers

What is a test set?

Answer: B. Real examples with expected answers

Examples plus correct answers.

- ✗ A. A list of tool features: no
- ✗ C. A collection of the tool's own past answers, saved so they can be shown to new users: no expected answers
- ✗ D. A price list: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S114-M-Q2 · question page 2376

AI.EVAL.01 · Build a small test set with expected answers

Who should determine the expected answers?

Answer: A. You, by working them out or confirming them

Independent answers.

- ✗ B. A random guess: no

✗ C. Nobody: needed

✗ D. The tool being tested, since it knows the correct answers better than any person could: circular

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S114-M-Q3 · question page 2377

AI.EVAL.01 · Build a small test set with expected answers

In this prepared example, how many did the tool get right?

Prepared example (a fictional five-item slice of a test set)

Email	Expected category	Tool's category
1	Billing	Billing
2	Shipping	Shipping
3	Billing	Refund
4	Refund	Refund
5	Shipping	Shipping

Answer: C. 4 of 5

Only email 3 is wrong.

✗ A. 2 of 5: miscounted

✗ B. 3 of 5: miscounted

✗ D. 5 of 5: they differ

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: none (stable)

S114-M-Q4 · question page 2378

AI.EVAL.01 · Build a small test set with expected answers

In the same example, what percent is correct?

Prepared example (a fictional five-item slice of a test set)

Email	Expected category	Tool's category
1	Billing	Billing
2	Shipping	Shipping
3	Billing	Refund
4	Refund	Refund
5	Shipping	Shipping

Answer: D. 80%

4 / 5 = 80%.

- ✗ A. 4%: arithmetic slip
- ✗ B. 40%: miscounted
- ✗ C. 100%: one wrong

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: none (stable)

S114-M-Q5 · question page 2379

AI.EVAL.01 · Build a small test set with expected answers

Which examples should a test set include?

Answer: B. Ordinary cases and a few hard ones

Realistic mix.

- ✗ A. Only made-up examples: use real ones
- ✗ C. Only one example: too few
- ✗ D. Only the easiest examples you can find: misleading

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S114-B1-Q1 · question page 2380

AI.EVAL.01 · Compare a test set with a practice exam with an answer key

What does the answer key do?

Answer: D. Lets you tell right from wrong

A trusted reference.

- ✗ A. Shortens the exam: no
- ✗ B. Grades students: not here
- ✗ C. Makes the textbook's answers correct: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S114-B1-Q2 · question page 2381

AI.EVAL.01 · Compare a test set with a practice exam with an answer key

Who writes the key in the example?

Answer: A. The teacher

Independent answers.

- ✗ B. A student: not here
- ✗ C. Nobody: the teacher
- ✗ D. The textbook: circular

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S114-B1-Q3 · question page 2382

AI.EVAL.01 · Compare a test set with a practice exam with an answer key

Why not use only easy problems?

Answer: C. You would learn little about hard cases

Include hard cases.

- ✗ A. Because easy problems always have wrong answers in every textbook that has been printed: no
- ✗ B. Because keys cannot be written for them: they can
- ✗ D. Because they take too long: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S114-B1-Q4 · question page 2383

AI.EVAL.01 · Compare a test set with a practice exam with an answer key

In AI terms, the 15 problems are like what?

Answer: A. A test set

Examples to test with.

- ✗ B. The tool's price: no
- ✗ C. A single question: no
- ✗ D. The tool's marketing materials, which show only examples that the tool handles well: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S114-B1-Q5 · question page 2384

AI.EVAL.01 · Compare a test set with a practice exam with an answer key

What can you not do without a key?

Answer: C. Tell right answers from wrong ones

No reference, no judgment.

- ✗ A. Turn the pages: no
- ✗ B. Buy the book: no
- ✗ D. Read the textbook's problems: you can read

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S114-F1-Q1 · question page 2385

AI.EVAL.01 · Build and use test sets in practice

A clinic wants to test a tool that drafts appointment reminders. What should its test set contain?

Answer: B. Real past appointments with the correct reminder details

Real examples, known answers.

- ✗ A. One appointment: too few
- ✗ C. Ten examples invented by the tool itself, along with the tool's own idea of the correct reminder: circular
- ✗ D. Only the clinic's logo: irrelevant

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S114-F1-Q2 · question page 2386

AI.EVAL.01 · Build and use test sets in practice

A tool gets 18 of 20 test emails right. What percent is that?

Answer: D. 90%

18 / 20 = 90%.

- ✗ A. 95%: arithmetic slip
- ✗ B. 80%: arithmetic slip
- ✗ C. 18%: arithmetic slip

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: none (stable)

S114-F1-Q3 · question page 2387

AI.EVAL.01 · Build and use test sets in practice

After the test, what should you write down?

Answer: C. Which examples failed and how

Failures teach most.

- ✗ A. Only the tool's name: not enough
- ✗ B. Only the final percent, since the details of each failure are not useful for decisions: lose insight

✗ D. Nothing: record results

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S114-F1-Q4 · question page 2388

AI.EVAL.01 · Build and use test sets in practice

Which example adds a useful hard case for a receipt reader?

Answer: A. A crumpled receipt with a faded total

Realistic difficulty.

✗ B. A blank page: not a receipt
✗ C. A photo of a cat: irrelevant
✗ D. A clean, printed receipt with large numbers: ordinary

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S114-F1-Q5 · question page 2389

AI.EVAL.01 · Build and use test sets in practice

Why use real examples rather than invented ones?

Answer: B. They reflect the task the tool will actually face

Realism.

✗ A. Because real examples are shorter: not the reason
✗ C. Because invented ones are illegal: no
✗ D. Because invented examples are always too difficult: not the reason

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S114-F2-Q1 · question page 2390

AI.EVAL.01 · Avoid test-set mistakes

A test set contains only easy examples. What happens?

Answer: D. The tool looks better than it is

Biased difficulty.

- ✗ A. The test cannot run: it runs
- ✗ B. Nothing changes: it misleads
- ✗ C. The tool looks worse than it really is, since easy examples confuse most AI tools: reversed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S114-F2-Q2 · question page 2391

AI.EVAL.01 · Avoid test-set mistakes

Your expected answer for one example is wrong. What might you conclude incorrectly?

Answer: B. That the tool made a mistake

Check your key.

- ✗ A. That the tool is faster: irrelevant
- ✗ C. That the example is real: irrelevant
- ✗ D. That the test set is perfect: it does

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S114-F2-Q3 · question page 2392

AI.EVAL.01 · Avoid test-set mistakes

You keep changing your request until all 20 examples pass. What is the risk?

Answer: A. It may fail on new examples

Keep some examples aside.

- ✗ B. The request becomes shorter: irrelevant

✗ C. The tool gets slower: irrelevant

✗ D. No risk: overfit

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S114-F2-Q4 · question page 2393

AI.EVAL.01 · Avoid test-set mistakes

When should you rerun the test set?

Answer: C. When the tool or your instructions change

Retest after changes.

✗ A. Only on holidays: no

✗ B. Never, since a test set only needs to be run one time when you first choose the tool: changes matter

✗ D. Every minute: excessive

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S114-F2-Q5 · question page 2394

AI.EVAL.01 · Avoid test-set mistakes

A real failure surprises you. What should you do with it?

Answer: D. Add it to the test set

Grow the test set.

✗ A. Delete the test set: no

✗ B. Hide it: no

✗ C. Ignore it: they can recur

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S115-M-Q1 · question page 2396

AI.EVAL.02 · Read false positives and false negatives and decide which matters more

What is a false positive?

Answer: A. A wrong "yes"

Flagged, but should not have been.

- ✗ B. A correct "no": true negative
- ✗ C. A correct "yes": true positive
- ✗ D. A wrong "no," where the tool missed something that it should have flagged as a yes: that is false negative

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S115-M-Q2 · question page 2397

AI.EVAL.02 · Read false positives and false negatives and decide which matters more

In this prepared example, how many false positives are there?

Prepared example (fictional results for 200 emails)

	Tool said spam	Tool said not spam
Really spam	45	5
Really not spam	10	140

Answer: D. 10

Real emails marked spam.

- ✗ A. 45: true positives
- ✗ B. 5, which is the number of real spam emails that the tool failed to mark as spam: that is false negatives
- ✗ C. 140: true negatives

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S115-M-Q3 · question page 2398

AI.EVAL.02 · Read false positives and false negatives and decide which matters more

In the same example, how many false negatives are there?

Prepared example (fictional results for 200 emails)

	Tool said spam	Tool said not spam
Really spam	45	5
Really not spam	10	140

Answer: B. 5

Spam let through.

- ✗ A. 140: true negatives
- ✗ C. 45: true positives
- ✗ D. 10: that is false positives

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S115-M-Q4 · question page 2399

AI.EVAL.02 · Read false positives and false negatives and decide which matters more

For a smoke detector, which mistake is worse?

Answer: C. A false negative

Missing a real fire.

- ✗ A. Neither matters: both matter
- ✗ B. They are always equal: not here
- ✗ D. A false positive: reversed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S115-M-Q5 · question page 2400

AI.EVAL.02 · Read false positives and false negatives and decide which matters more

What often happens when you reduce one kind of mistake?

Answer: B. The other kind may increase

A trade-off.

- ✗ A. The tool stops working: not necessarily
- ✗ C. Both kinds of mistake always disappear completely at the same time without any trade-off: usually a trade-off
- ✗ D. Nothing changes: it changes

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S115-B1-Q1 · question page 2401

AI.EVAL.02 · Read a 2-by-2 results table

How many true positives?

Answer: D. 30

Damaged and flagged.

- ✗ A. 4: false positives
- ✗ B. 6, since those parcels were really damaged and the tool said something about them: false negatives
- ✗ C. 160: true negatives

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S115-B1-Q2 · question page 2402

AI.EVAL.02 · Read a 2-by-2 results table

How many false negatives?

Answer: A. 6

Damaged but missed.

- ✗ B. 30: true positives
- ✗ C. 4, because those parcels were flagged by the tool even though they were not damaged: false positives
- ✗ D. 160: true negatives

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S115-B1-Q3 · question page 2403

AI.EVAL.02 · Read a 2-by-2 results table

How many false positives?

Answer: C. 4

Flagged but fine.

- ✗ A. 30: true positives
- ✗ B. 6, which counts the damaged parcels that the tool failed to flag before shipping: false negatives
- ✗ D. 160: true negatives

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S115-B1-Q4 · question page 2404

AI.EVAL.02 · Read a 2-by-2 results table

How many parcels in total?

Answer: D. 200

30 + 6 + 4 + 160 = 200.

- ✗ A. 166: arithmetic slip
- ✗ B. 190: arithmetic slip

✗ C. 36: only damaged

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: none (stable)

S115-B1-Q5 · question page 2405

AI.EVAL.02 · Read a 2-by-2 results table

How many were really damaged?

Answer: B. 36

30 + 6 = 36.

✗ A. 200: total

✗ C. 34, the number of parcels the tool flagged as damaged, whether or not they really were: that is flagged

✗ D. 30: missed some

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: none (stable)

S115-F1-Q1 · question page 2406

AI.EVAL.02 · Decide which error matters more in practice

A tool screens job applications and wrongly rejects qualified people. Which error is this?

Answer: A. A false negative

Qualified people missed.

✗ B. A true negative: not correct

✗ C. A true positive: not correct

✗ D. A false positive: it said no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S115-F1-Q2 · question page 2407

AI.EVAL.02 · Decide which error matters more in practice

A medical screening tool flags patients for a follow-up test. Which error is usually worse?

Answer: C. Missing someone who is sick

False negatives can be dangerous.

- ✗ A. Both are always equal: not usually
- ✗ B. Neither: both matter
- ✗ D. Flagging a healthy person: usually reversed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S115-F1-Q3 · question page 2408

AI.EVAL.02 · Decide which error matters more in practice

A spam filter hides a job offer email. Which error is this?

Answer: A. A false positive

A real email marked spam.

- ✗ B. A true negative: it was flagged
- ✗ C. A true positive: not spam
- ✗ D. A false negative: it said spam

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S115-F1-Q4 · question page 2409

AI.EVAL.02 · Decide which error matters more in practice

Results: 90 real fraud cases caught, 10 missed. How many false negatives?

Answer: B. 10

Missed cases.

- ✗ A. 100: total cases
- ✗ C. 0: some missed

✗ D. 90: those were caught

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S115-F1-Q5 · question page 2410

AI.EVAL.02 · Decide which error matters more in practice

For a library's "overdue book" reminder, a false positive sends a reminder for a returned book. How serious is it?

Answer: D. Mildly annoying, usually

Low-stakes error.

✗ A. Worse than any missed reminder: not usually
✗ B. Impossible: it can happen
✗ C. Extremely dangerous: overstated

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S115-F2-Q1 · question page 2411

AI.EVAL.02 · Choose and explain the error balance

A reading-help screener accepts more false positives. Why might that be reasonable?

Answer: C. Missing a student who needs help is worse

Prioritize catching needs.

✗ A. Because it saves money always: not necessarily
✗ B. Because screeners must be wrong: no
✗ D. Because false positives never have any cost: they cost a little

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S115-F2-Q2 · question page 2412

AI.EVAL.02 · Choose and explain the error balance

A fraud flagger should have very few false positives. Why?

Answer: D. A wrong flag harms an innocent person

Protect the innocent.

- ✗ A. Because false negatives do not exist: they do
- ✗ B. Because flags are expensive to print: not the reason
- ✗ C. Because fraud is never real: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S115-F2-Q3 · question page 2413

AI.EVAL.02 · Choose and explain the error balance

Is choosing the balance a technical fact or a decision?

Answer: B. A decision people must make

Values guide it.

- ✗ A. A technical fact that is fixed by the model and that nobody can choose or change: it is a choice
- ✗ C. Neither: it is a decision
- ✗ D. A law of physics: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S115-F2-Q4 · question page 2414

AI.EVAL.02 · Choose and explain the error balance

What should a team say when setting a tool's threshold?

Answer: A. Which error it is limiting and why

Transparency.

- ✗ B. Only the tool's name: not enough

✗ C. Only the date: not enough

✗ D. Nothing, since the threshold setting is a private technical detail that nobody needs to know: explain

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S115-F2-Q5 · question page 2415

AI.EVAL.02 · Choose and explain the error balance

A smoke detector maker lowers false alarms but now misses more fires. Is this a good trade?

Answer: C. No; missed fires are far worse

Weigh the costs.

✗ A. Yes, because people find false alarms annoying, and annoyance is the most important factor: safety first

✗ B. It does not matter: it matters

✗ D. Yes, always: not here

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S116-M-Q1 · question page 2417

AI.EVAL.03 · Explain what a benchmark score shows and its limits

What is a benchmark?

Answer: A. A standard test used to compare models

Shared test.

✗ B. A model's price: no

✗ C. A company's ranking of its own staff: no

✗ D. A long bench in a computer lab: not the meaning

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: eval_contamination

S116-M-Q2 · question page 2418

AI.EVAL.03 · Explain what a benchmark score shows and its limits

What does a high benchmark score show?

Answer: D. Good performance on that test

Specific to the test.

- ✗ A. That it is cheap: no
- ✗ B. That the model is conscious: no
- ✗ C. Good performance on every task: too broad

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: eval_contamination

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S116-M-Q3 · question page 2419

AI.EVAL.03 · Explain what a benchmark score shows and its limits

What is contamination?

Answer: C. Test questions leaking into training data

Seen-before answers.

- ✗ A. A model refusing a test: no
- ✗ B. A virus that infects the computers used to train models, making them run more slowly: not the meaning
- ✗ D. Dust in a data center: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: eval_contamination

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S116-M-Q4 · question page 2420

AI.EVAL.03 · Explain what a benchmark score shows and its limits

How can contamination affect scores?

Answer: B. It can inflate them

Memorized answers look like skill.

- ✗ A. It has no effect at all: it can inflate
- ✗ C. It makes scores random: not the finding
- ✗ D. It always lowers scores: reversed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: eval_contamination

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S116-M-Q5 · question page 2421

AI.EVAL.03 · Explain what a benchmark score shows and its limits

After reading benchmark scores, what should you do?

Answer: C. Test on your own examples

Your task decides.

- ✗ A. Ask the vendor: biased
- ✗ B. Ignore all scores: they help a little
- ✗ D. Choose the model with the highest score and use it without any further testing at all: may not fit

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: eval_contamination

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S116-B1-Q1 · question page 2422

AI.EVAL.03 · Compare a benchmark with a driving test

What does passing a driving test show?

Answer: A. You can do the tested skills on that day

Limited proof.

- ✗ B. That you are a mechanic: no
- ✗ C. That you own a car: no
- ✗ D. That you can drive perfectly in every condition anywhere in the world for the rest of your life: too broad

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: eval_contamination

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S116-B1-Q2 · question page 2423

AI.EVAL.03 · Compare a benchmark with a driving test

Driving in a snowstorm is like what for a model?

Answer: B. A task different from the benchmark

Untested conditions.

- ✗ A. A faster computer: no
- ✗ C. A price change: no
- ✗ D. The exact benchmark questions: opposite

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: eval_contamination

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S116-B1-Q3 · question page 2424

AI.EVAL.03 · Compare a benchmark with a driving test

Memorizing the exact route is like what?

Answer: D. Contamination

Seen answers.

- ✗ A. A false negative: no
- ✗ B. A subscription: no
- ✗ C. A fair test: not fair

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: eval_contamination

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S116-B1-Q4 · question page 2425

AI.EVAL.03 · Compare a benchmark with a driving test

Why does memorizing the route weaken the test?

Answer: A. The pass shows less real skill

Inflated result.

- ✗ B. Because memorizing routes makes cars drive more slowly during the test itself: no
- ✗ C. Because testers object: not the point
- ✗ D. Because routes change daily: not the point

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: eval_contamination

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S116-B1-Q5 · question page 2426

AI.EVAL.03 · Compare a benchmark with a driving test

What is the lesson about benchmarks?

Answer: B. Useful but limited to what they test

Rough guide.

- ✗ A. They are about cars: no
- ✗ C. They are worthless and should never be looked at by anyone choosing a model: too strong
- ✗ D. They prove everything: too strong

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: eval_contamination

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S116-F1-Q1 · question page 2427

AI.EVAL.03 · Interpret benchmark claims in practice

A model scores highly on an English reading benchmark. You need it for Vietnamese customer emails. What should you do?

Answer: C. Test it on real Vietnamese emails

Your task differs.

- ✗ A. Ignore the language: it matters

✗ B. Choose a model with a lower English score: not the method

✗ D. Use it right away, since high English scores always mean equally strong results in every language: not guaranteed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: eval_contamination

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S116-F1-Q2 · question page 2428

AI.EVAL.03 · Interpret benchmark claims in practice

A model's math score jumps suddenly, and the test questions were posted online years ago. What should you suspect?

Answer: D. Possible contamination

Leaked questions inflate scores.

✗ A. That the model must have become truly brilliant at mathematics overnight without any other explanation: consider alternatives

✗ B. That the score is lower: it rose

✗ C. That math changed: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: eval_contamination

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S116-F1-Q3 · question page 2429

AI.EVAL.03 · Interpret benchmark claims in practice

An ad lists three benchmarks where its model wins. What question should you ask?

Answer: B. "What about other benchmarks?"

Selective reporting.

✗ A. "Who designed the logo?": irrelevant

✗ C. "How many colors are used in the chart that shows the three benchmark scores in the ad?": irrelevant

✗ D. "Is the ad long?": irrelevant

Arithmetic checked by code: not applicable. Code example run and its output checked: not

applicable. Sources: eval_contamination

S116-F1-Q4 · question page 2430

AI.EVAL.03 · Interpret benchmark claims in practice

Which benchmark is most relevant for a tool that summarizes legal contracts?

Answer: D. One using real contract summaries

Match the task.

- ✗ A. A benchmark of grade-school arithmetic word problems, since all tasks use the same skills: unrelated
- ✗ B. A trivia quiz: unrelated
- ✗ C. A poetry benchmark: unrelated

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: eval_contamination

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S116-F1-Q5 · question page 2431

AI.EVAL.03 · Interpret benchmark claims in practice

Two models score 91 and 92 on a benchmark. What is reasonable?

Answer: C. The difference may be noise; test both

Small gaps may not matter.

- ✗ A. Neither works: not shown
- ✗ B. Model A is better: not shown
- ✗ D. Model B is clearly far better in every way: over-reading

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: eval_contamination

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S116-F2-Q1 · question page 2432

AI.EVAL.03 · Avoid over-reading benchmark scores

All top models score about 99 on a benchmark. What does this mean?

Answer: A. It no longer separates them well

Saturation.

- ✗ B. That scores went down: they are high
- ✗ C. That the benchmark is broken: not necessarily
- ✗ D. That every model is now perfect: too broad

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: eval_contamination

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S116-F2-Q2 · question page 2433

AI.EVAL.03 · Avoid over-reading benchmark scores

Which question best checks a benchmark's relevance?

Answer: D. "Is this test like my task?"

Similarity to your use.

- ✗ A. "Is the model new?": not the key
- ✗ B. "Is the score a round number?": irrelevant
- ✗ C. "How many people shared the benchmark results on social media during the first week?": irrelevant

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: eval_contamination

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S116-F2-Q3 · question page 2434

AI.EVAL.03 · Avoid over-reading benchmark scores

Who ran a benchmark matters because...

Answer: B. Methods and incentives can affect results

Source and method.

- ✗ A. It does not matter: it does
- ✗ C. Benchmarks run themselves: no
- ✗ D. Only famous people are allowed to run benchmarks: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: eval_contamination

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S116-F2-Q4 · question page 2435

AI.EVAL.03 · Avoid over-reading benchmark scores

Why are simple overlap checks not enough to rule out contamination?

Answer: C. They can miss contaminated examples

Hard to detect.

- ✗ A. Because overlap checks always find every single leaked question with no mistakes: they can miss
- ✗ B. Because contamination does not exist: it does
- ✗ D. Because they are too slow: not the issue

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: eval_contamination

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S116-F2-Q5 · question page 2436

AI.EVAL.03 · Avoid over-reading benchmark scores

What is the final step after reading benchmarks?

Answer: A. Check with your own small test set

Your evidence.

- ✗ B. Buy the tool with the most benchmark wins and never test it on any of your own tasks: skip testing
- ✗ C. Ask friends only: not enough
- ✗ D. Read more ads: not a check

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: eval_contamination

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S117-M-Q1 · question page 2438

AI.EVAL.04 · Weigh accuracy against speed and cost for a stated need

What does accuracy measure here?

Answer: B. How often the tool is right

Correctness rate.

- ✗ A. How long answers are: no
- ✗ C. How much it costs: that is cost
- ✗ D. How quickly the answer appears: that is speed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S117-M-Q2 · question page 2439

AI.EVAL.04 · Weigh accuracy against speed and cost for a stated need

Which task needs accuracy above all?

Answer: C. Checking medicine instructions

High stakes.

- ✗ A. Brainstorming party themes: low stakes
- ✗ B. Sorting old photos: low stakes
- ✗ D. Suggesting fun names: low stakes

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S117-M-Q3 · question page 2440

AI.EVAL.04 · Weigh accuracy against speed and cost for a stated need

In this example, which option meets "at least 95% correct" at the lowest cost?

Prepared example (fictional measurements and hypothetical prices)

Option	Correct on test set	Seconds per answer	Cost per 1,000 tasks (hypothetical)
P	98%	6	20 dollars
Q	95%	2	5 dollars

R	85%	1	1 dollar

Answer: A. Q

Q is 95% at 5 dollars.

✗ B. P, because the most accurate option is always the cheapest choice that meets any minimum: costs more

✗ C. R: below 95%

✗ D. None: Q meets it

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S117-M-Q4 · question page 2441

AI.EVAL.04 · Weigh accuracy against speed and cost for a stated need

In the same example, which option is fastest?

Prepared example (fictional measurements and hypothetical prices)

Option	Correct on test set	Seconds per answer	Cost per 1,000 tasks (hypothetical)
P	98%	6	20 dollars
Q	95%	2	5 dollars
R	85%	1	1 dollar

Answer: D. R

1 second per answer.

✗ A. They are equal: they differ

✗ B. Q: 2 seconds

✗ C. P, since the most accurate option always gives its answers in the shortest time: slowest

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S117-M-Q5 · question page 2442

AI.EVAL.04 · Weigh accuracy against speed and cost for a stated need

What is the suggested method?

Answer: C. Set a minimum for what matters most, then pick the cheapest that meets it

Minimum, then optimize.

- ✗ A. Choose the fastest no matter what: ignores needs
- ✗ B. Choose at random: no
- ✗ D. Always choose the most expensive option, since price is the best measure of overall quality: not the method

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S117-B1-Q1 · question page 2443

AI.EVAL.04 · Compare the trade-off with choosing a delivery service

Overnight delivery is like what in AI tools?

Answer: B. A fast but expensive option

Speed costs more.

- ✗ A. A benchmark: no
- ✗ C. A free plan: no
- ✗ D. A slow and very cheap option: that is economy

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S117-B1-Q2 · question page 2444

AI.EVAL.04 · Compare the trade-off with choosing a delivery service

For a box of old books, what matters most?

Answer: D. Low cost

Cheap is fine.

- ✗ A. Tracking every minute: not needed
- ✗ B. Gift wrap: no
- ✗ C. Overnight speed: not needed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S117-B1-Q3 · question page 2445

AI.EVAL.04 · Compare the trade-off with choosing a delivery service

For a passport application, what matters most?

Answer: A. Reliability

Like accuracy.

- ✗ B. Nothing: it matters
- ✗ C. The color of the envelope: irrelevant
- ✗ D. The cheapest possible price: risky

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S117-B1-Q4 · question page 2446

AI.EVAL.04 · Compare the trade-off with choosing a delivery service

What decides the choice?

Answer: D. What the package needs

Task needs.

- ✗ A. Whichever option is advertised most often on television and online that week: irrelevant
- ✗ B. The sender's mood: no
- ✗ C. The weather: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S117-B1-Q5 · question page 2447

AI.EVAL.04 · Compare the trade-off with choosing a delivery service

What is the lesson for AI tools?

Answer: B. Let the task decide among accuracy, speed and cost

Trade-offs.

- ✗ A. Never use tools: no
- ✗ C. Always pick the cheapest: not always
- ✗ D. Always pick the fastest tool, because speed matters more than anything else for every task: not always

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S117-F1-Q1 · question page 2448

AI.EVAL.04 · Make accuracy, speed and cost decisions in practice

Using the earlier example, a team needs answers in under 3 seconds and at least 90% correct. Which option fits?

Prepared example (fictional measurements and hypothetical prices)

Option	Correct on test set	Seconds per answer	Cost per 1,000 tasks (hypothetical)
P	98%	6	20 dollars
Q	95%	2	5 dollars
R	85%	1	1 dollar

Answer: A. Q

Q: 95%, 2 seconds.

- ✗ B. P, because it is the most accurate, and accuracy matters more than any

time limit: 6 seconds

✗ C. None: Q fits

✗ D. R: below 90%

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S117-F1-Q2 · question page 2449

AI.EVAL.04 · Make accuracy, speed and cost decisions in practice

A school sorts 10,000 old forms overnight; errors are easy to fix. Which factor matters most?

Answer: C. Low cost

Batch, low stakes.

✗ A. Speed, because the forms must be sorted within one second each, even though it runs overnight: overnight

✗ B. Perfect accuracy at any price: errors are easy to fix

✗ D. None: cost matters

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S117-F1-Q3 · question page 2450

AI.EVAL.04 · Make accuracy, speed and cost decisions in practice

A pharmacy helper answers dosage questions. Which factor matters most?

Answer: A. Accuracy

High stakes.

✗ B. Speed above all: not above accuracy

✗ C. Answer length: no

✗ D. Low cost: safety first

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S117-F1-Q4 · question page 2451

AI.EVAL.04 · Make accuracy, speed and cost decisions in practice

Hypothetical: option R costs 1 dollar per 1,000 tasks and Q costs 5. For 20,000 tasks, how much more is Q?

Answer: C. 80 dollars

20 x 5 - 20 x 1 = 80.

- ✗ A. 100 dollars: that is Q total
- ✗ B. 20 dollars: that is R total
- ✗ D. 4 dollars: scale up

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: none (stable)

S117-F1-Q5 · question page 2452

AI.EVAL.04 · Make accuracy, speed and cost decisions in practice

Which is the right first step?

Answer: D. Decide what the task needs

Needs first.

- ✗ A. Pick the cheapest tool: not yet
- ✗ B. Pick the newest tool: not the method
- ✗ C. Pick the tool: backward

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S117-F2-Q1 · question page 2453

AI.EVAL.04 · Include hidden costs in the comparison

Option R makes 150 errors per 1,000 tasks and Q makes 50. Each error takes 10 minutes to fix. How much more fixing time does R need per 1,000 tasks?

Answer: B. 1,000 minutes

(150 - 50) x 10 = 1,000.

- ✗ A. 1,500 minutes: that is R total
- ✗ C. 500 minutes: arithmetic slip
- ✗ D. 100 minutes: ten minutes each

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: none (stable)

S117-F2-Q2 · question page 2454

AI.EVAL.04 · Include hidden costs in the comparison

What does the previous answer show?

Answer: C. A cheaper tool can cost more overall

Hidden costs.

- ✗ A. That Q is slower: not shown
- ✗ B. That errors do not matter: they do
- ✗ D. That fixing errors is always free: people's time costs

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S117-F2-Q3 · question page 2455

AI.EVAL.04 · Include hidden costs in the comparison

A slow but accurate tool is used for overnight reports. Is slowness a problem?

Answer: A. Probably not, since no one is waiting

Context matters.

- ✗ B. Yes, because reports must be instant: not overnight
- ✗ C. It cannot be judged: it can
- ✗ D. Yes: depends

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S117-F2-Q4 · question page 2456

AI.EVAL.04 · Include hidden costs in the comparison

What should a fair comparison include besides the tool's price?

Answer: D. Time people spend checking and fixing

Full cost.

- ✗ A. The tool's logo: irrelevant
- ✗ B. The tool's age: irrelevant
- ✗ C. Only the price of the tool, since people's time is never counted as a cost in practice: it counts

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S117-F2-Q5 · question page 2457

AI.EVAL.04 · Include hidden costs in the comparison

Prices drop and a new option appears. What should you do?

Answer: B. Revisit the choice with your test set

Re-evaluate.

- ✗ A. Ignore the news: may miss value
- ✗ C. Switch immediately without testing: risky
- ✗ D. Keep the old choice forever: not always

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S118-M-Q1 · question page 2459

AI.EVAL.05 · Compare two versions with the same tests and written criteria

What makes a comparison fair?

Answer: C. Same tests and criteria written beforehand

Same tests, prior criteria.

- ✗ A. Judging by one memorable answer: unreliable
- ✗ B. Criteria written after seeing results: biased
- ✗ D. Different tests for each version: unfair

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S118-M-Q2 · question page 2460

AI.EVAL.05 · Compare two versions with the same tests and written criteria

Why change only one thing at a time?

Answer: A. To know what caused any difference

Isolate the cause.

- ✗ B. Because changing several things at once is not allowed by any of the AI providers: not the reason
- ✗ C. To make tests longer: no
- ✗ D. To save money: not the reason

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S118-M-Q3 · question page 2461

AI.EVAL.05 · Compare two versions with the same tests and written criteria

In this example, how many does Version B get right?

Prepared example (fictional side-by-side results on the same six tests)

Example	Version A	Version B
1	Correct	Correct
2	Wrong	Correct
3	Correct	Wrong
4	Correct	Correct

5	Wrong	Correct
6	Correct	Correct

Answer: D. 5 of 6

Only example 3 is wrong.

- ✗ A. 3 of 6: miscounted
- ✗ B. 6 of 6: one wrong
- ✗ C. 4 of 6: that is A

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: none (stable)

S118-M-Q4 · question page 2462

AI.EVAL.05 · Compare two versions with the same tests and written criteria

In the same example, which test shows a regression for Version B?

Prepared example (fictional side-by-side results on the same six tests)

Example	Version A	Version B
1	Correct	Correct
2	Wrong	Correct
3	Correct	Wrong
4	Correct	Correct
5	Wrong	Correct
6	Correct	Correct

Answer: B. Example 3

A worked, B broke it.

- ✗ A. Example 5: a fix
- ✗ C. Example 1: both correct
- ✗ D. Example 2: that is a fix

Arithmetic checked by code: not applicable. Code example run and its output checked: not

applicable. Sources: none (stable)

S118-M-Q5 · question page 2463

AI.EVAL.05 · Compare two versions with the same tests and written criteria

Why hide which version produced each answer when judging?

Answer: D. So expectations do not bias the judgment

Blind judging.

- ✗ A. Because answers have no labels: they can
- ✗ B. To save time: not the reason
- ✗ C. Because version names are secret information: not the reason

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S118-B1-Q1 · question page 2464

AI.EVAL.05 · Compare version testing with a fair taste test

Why use the same oven for both recipes?

Answer: B. So only the recipe differs

One change at a time.

- ✗ A. To save electricity: not the point
- ✗ C. Because ovens can only be used once each day and must be shared by both recipes: no
- ✗ D. Because ovens are expensive: not the point

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S118-B1-Q2 · question page 2465

AI.EVAL.05 · Compare version testing with a fair taste test

Deciding "taste, texture, appearance" in advance is like what?

Answer: A. Writing criteria beforehand

Prior criteria.

- ✗ B. Choosing what "better" means after tasting, so that your favorite recipe can win easily: opposite
- ✗ C. Buying flour: no
- ✗ D. Using the same tasters: related but different

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S118-B1-Q3 · question page 2466

AI.EVAL.05 · Compare version testing with a fair taste test

Tasters not knowing which is which is like what?

Answer: C. Blind judging

Hide the labels.

- ✗ A. Baking twice: no
- ✗ B. Using one oven: no
- ✗ D. Telling the tasters: opposite

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S118-B1-Q4 · question page 2467

AI.EVAL.05 · Compare version testing with a fair taste test

Changing recipe and oven at once causes what problem?

Answer: D. You cannot tell which change mattered

Confounded changes.

- ✗ A. The cookies always burn, since two changes at once are too much for any oven: not the point
- ✗ B. No problem: there is one

✗ C. The cookies get bigger: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S118-B1-Q5 · question page 2468

AI.EVAL.05 · Compare version testing with a fair taste test

What is the lesson for AI versions?

Answer: C. Same tests, prior criteria, one change, blind judging

Fair comparison.

✗ A. Never compare: no

✗ B. Test each version on different examples so that both versions get a fair chance to shine: unfair

✗ D. Taste only the new version: no comparison

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S118-F1-Q1 · question page 2469

AI.EVAL.05 · Run fair version comparisons in practice

Ana changes both her prompt and the model, and results improve. What can she conclude?

Answer: A. Not which change helped

Two changes at once.

✗ B. Nothing improved: it improved

✗ C. That the new model caused all of the improvement, since models matter more than prompts: unknown

✗ D. That the prompt caused it: unknown

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S118-F1-Q2 · question page 2470

AI.EVAL.05 · Run fair version comparisons in practice

A team tests the new version on 3 examples and the old on 20 different ones. What is wrong?

Answer: B. Different tests make the comparison unfair

Same tests for both.

- ✗ A. The old version needs more tests: not the main issue
- ✗ C. The new version needs fewer: no
- ✗ D. Nothing, because a small number of examples is always enough for any new version of a tool: unfair

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S118-F1-Q3 · question page 2471

AI.EVAL.05 · Run fair version comparisons in practice

After seeing results, Raj decides "shorter answers are better," which favors his choice. What is the problem?

Answer: D. Criteria chosen after results can be biased

Write criteria first.

- ✗ A. Shorter answers are always better: not the point
- ✗ B. Criteria do not matter: they do
- ✗ C. No problem, since criteria should always be chosen after seeing which version is better: biased

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S118-F1-Q4 · question page 2472

AI.EVAL.05 · Run fair version comparisons in practice

With only 4 test examples, one changed answer moves the score by how much?

Answer: B. 25 percentage points

1 / 4 = 25%.

- ✗ A. 50 percentage points: no
- ✗ C. 4 percentage points: no
- ✗ D. 1 percentage point: small sample

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: none (stable)

S118-F1-Q5 · question page 2473

AI.EVAL.05 · Run fair version comparisons in practice

Which practice reduces bias when judging writing quality?

Answer: C. Hiding which version wrote each answer

Blind review.

- ✗ A. Letting the developer judge alone: biased
- ✗ B. Judging only the best answer: cherry-picking
- ✗ D. Telling judges: adds bias

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S118-F2-Q1 · question page 2474

AI.EVAL.05 · Catch unfair comparisons and regressions

What is a regression?

Answer: A. Something that used to work now fails

Broken old behavior.

- ✗ B. A price increase: no
- ✗ C. A slower computer: no
- ✗ D. A new feature that works better: opposite

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S118-F2-Q2 · question page 2475

AI.EVAL.05 · Catch unfair comparisons and regressions

A new version fixes 5 problems but breaks 4 that worked before. What should you report?

Answer: B. Both the fixes and the regressions

Full picture.

- ✗ A. Nothing: report results
- ✗ C. Only the regressions: incomplete
- ✗ D. Only the five fixes, since regressions are normal and do not need to be mentioned to anyone: hides problems

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S118-F2-Q3 · question page 2476

AI.EVAL.05 · Catch unfair comparisons and regressions

Someone judges a new model by one impressive answer they remember. What is wrong?

Answer: A. One answer is not a fair test

Use the full test set.

- ✗ B. Memories are always wrong: not the point
- ✗ C. The answer was too short: not the point
- ✗ D. Nothing: not proof

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S118-F2-Q4 · question page 2477

AI.EVAL.05 · Catch unfair comparisons and regressions

The new version was tested only on easy examples. What can result?

Answer: C. An overly positive comparison

Biased examples.

- ✗ A. No result: there is a result
- ✗ B. A perfectly fair comparison: unfair
- ✗ D. An overly negative comparison: reversed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S118-F2-Q5 · question page 2478

AI.EVAL.05 · Catch unfair comparisons and regressions

What protects against picking criteria that favor your preference?

Answer: D. Writing criteria before seeing results

Commit in advance.

- ✗ A. Using no criteria: no standard
- ✗ B. Changing criteria after each test: biased
- ✗ C. Choosing criteria together with the developer: biased

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S119-M-Q1 · question page 2480

AI.EVAL.06 · Use a rubric for human review and look for failures

What is a rubric?

Answer: C. A list of criteria with levels for scoring

Structured criteria.

- ✗ A. A type of model: no
- ✗ B. A single overall impression of whether an answer feels good or bad to the reviewer: not structured
- ✗ D. A price chart: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not

applicable. Sources: none (stable)

S119-M-Q2 · question page 2481

AI.EVAL.06 · Use a rubric for human review and look for failures

In this prepared example, what is the total score out of 8?

Prepared example (a fictional rubric score for one AI-written reminder)

Criterion	Score (0-2)	Reviewer note
Accurate	1	One date is wrong
Complete	2	All key points present
Clear	2	Easy to follow
Appropriate	1	Too long for a text message

Answer: A. 6

1 + 2 + 2 + 1 = 6.

✗ B. 4: miscounted

✗ C. 5: miscounted

✗ D. 8, because every criterion was at least partly met, which counts as a full score: two lost points

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: none (stable)

S119-M-Q3 · question page 2482

AI.EVAL.06 · Use a rubric for human review and look for failures

In the same example, which criterion found a factual problem?

Prepared example (a fictional rubric score for one AI-written reminder)

Criterion	Score (0-2)	Reviewer note
Accurate	1	One date is wrong

Complete	2	All key points present
Clear	2	Easy to follow
Appropriate	1	Too long for a text message

Answer: B. Accurate

A wrong date.

- ✗ A. Complete: scored 2
- ✗ C. Clear: scored 2
- ✗ D. Appropriate: that is tone and length

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S119-M-Q4 · question page 2483

AI.EVAL.06 · Use a rubric for human review and look for failures

Why be wary of demonstrations?

Answer: D. They show chosen examples that work

Selected successes.

- ✗ A. Because they are free: irrelevant
- ✗ B. Because demonstrations always use fake tools that do not exist in real life: not the reason
- ✗ C. Because they are too long: not the reason

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S119-M-Q5 · question page 2484

AI.EVAL.06 · Use a rubric for human review and look for failures

What does good evaluation look for on purpose?

Answer: C. Failures

Seek problems.

- ✗ A. Only short answers: not the goal
- ✗ B. Only the fastest answers: not the goal
- ✗ D. Only successes: biased

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S119-B1-Q1 · question page 2485

AI.EVAL.06 · Score an output with a rubric

The note names the wrong gift. What is the Accurate score?

Answer: B. 0

Not met.

- ✗ A. 1: it is not partly right
- ✗ C. 2, since naming a gift at all shows that the note is generally accurate: wrong gift
- ✗ D. 3: max is 2

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S119-B1-Q2 · question page 2486

AI.EVAL.06 · Score an output with a rubric

The note is friendly. What is the Warm score?

Answer: D. 2

Fully met.

- ✗ A. 1: fully met
- ✗ B. 0, because a friendly tone cannot make up for naming the wrong gift in the note: separate criteria
- ✗ C. 3: max is 2

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S119-B1-Q3 · question page 2487

AI.EVAL.06 · Score an output with a rubric

The note has 60 words. What is the Brief score?

Answer: A. 2

60 is under 80.

- ✗ B. 3: max is 2
- ✗ C. 1, since sixty words is quite long for a thank-you note, even though it is under eighty: meets limit
- ✗ D. 0: meets limit

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: none (stable)

S119-B1-Q4 · question page 2488

AI.EVAL.06 · Score an output with a rubric

What is the total?

Answer: C. 4 of 6

0 + 2 + 2 = 4.

- ✗ A. 5 of 6: miscounted
- ✗ B. 2 of 6: miscounted
- ✗ D. 6 of 6: accuracy failed

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: none (stable)

S119-B1-Q5 · question page 2489

AI.EVAL.06 · Score an output with a rubric

What should accompany each score?

Answer: B. A short reason

Explain the score.

- ✗ A. A drawing: no
- ✗ C. Nothing: reasons help
- ✗ D. A long essay: too much

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S119-F1-Q1 · question page 2490

AI.EVAL.06 · Use rubrics and failure-seeking in practice

A vendor demo shows five perfect summaries. What should a buyer do?

Answer: A. Test it on hard, real documents

Seek failures.

- ✗ B. Ask for the same five again: no new information
- ✗ C. Trust the vendor: biased
- ✗ D. Buy it immediately: chosen examples

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S119-F1-Q2 · question page 2491

AI.EVAL.06 · Use rubrics and failure-seeking in practice

Which rubric criterion is most specific?

Answer: D. "Includes the deadline and the fee"

Specific and checkable.

- ✗ A. "Is nice": vague
- ✗ B. "Feels right": vague
- ✗ C. "Is good quality overall and generally pleasant to read for most people who see it": vague

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S119-F1-Q3 · question page 2492

AI.EVAL.06 · Use rubrics and failure-seeking in practice

Two reviewers score the same letter 2 and 0 on "Clear." What does this suggest?

Answer: A. The criterion may be unclear

Refine the criterion.

- ✗ B. That rubrics never work: too strong
- ✗ C. That the letter is perfect: no
- ✗ D. That one of the two reviewers is lying about the score that they gave to the letter: not implied

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S119-F1-Q4 · question page 2493

AI.EVAL.06 · Use rubrics and failure-seeking in practice

Which test case seeks failure for a meeting summarizer?

Answer: D. A meeting where people disagree and change plans

Hard case.

- ✗ A. A meeting about the weather: easy case
- ✗ B. A one-sentence meeting: easy case
- ✗ C. A short meeting where one person reads an agenda: easy case

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S119-F1-Q5 · question page 2494

AI.EVAL.06 · Use rubrics and failure-seeking in practice

A reviewer scores 50 outputs in one sitting and later scores become generous. What helps?

Answer: B. Shorter review sessions

Avoid fatigue.

- ✗ A. Removing the rubric: worse
- ✗ C. Scoring without reading: worse
- ✗ D. Longer sessions: fatigue

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S119-F2-Q1 · question page 2495

AI.EVAL.06 · Keep rubric review reliable

Why have two reviewers score the same outputs?

Answer: C. To reveal unclear criteria

Check agreement.

- ✗ A. To avoid writing criteria: no
- ✗ B. To make the review take twice as long, which always improves the final results: not the reason
- ✗ D. To hide disagreements: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S119-F2-Q2 · question page 2496

AI.EVAL.06 · Keep rubric review reliable

What helps new reviewers score consistently?

Answer: D. Examples of each score level

Calibration examples.

- ✗ A. Only the final total: not enough
- ✗ B. Telling them to score however they feel, without any examples or written criteria: inconsistent

✗ C. Nothing: examples help

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S119-F2-Q3 · question page 2497

AI.EVAL.06 · Keep rubric review reliable

Why do vague criteria cause problems?

Answer: B. Reviewers interpret them differently

Inconsistent scoring.

✗ A. Because they cannot be written: they can
✗ C. Because they are always too strict: not necessarily
✗ D. Because vague criteria are shorter: not the problem

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S119-F2-Q4 · question page 2498

AI.EVAL.06 · Keep rubric review reliable

Reviewers disagree often. What should you do?

Answer: C. Refine the rubric

Clarify criteria.

✗ A. Fire a reviewer: not the fix
✗ B. Stop reviewing: no
✗ D. Average their scores: hides problem

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S119-F2-Q5 · question page 2499

AI.EVAL.06 · Keep rubric review reliable

Why does reviewer fatigue matter?

Answer: A. Tired reviewers score more leniently

Consistency drops.

- ✗ B. It only affects speed: also accuracy
- ✗ C. It does not matter: it does
- ✗ D. Tired reviewers always score more harshly: often lenient

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S120-M-Q1 · question page 2501

AI.EVAL.07 · Choose a measure that matches the goal of a task

What should a measure match?

Answer: B. The goal of the task

Goal-aligned.

- ✗ A. The tool's price: not the goal
- ✗ C. The tool's name: no
- ✗ D. Whatever number is easiest: misleading

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S120-M-Q2 · question page 2502

AI.EVAL.07 · Choose a measure that matches the goal of a task

Only 2 of 100 parcels are damaged. A tool always says "fine." What percent is it correct?

Answer: D. 98%

98 of 100 are fine.

- ✗ A. 2%: reversed
- ✗ B. 100%: misses 2

✗ C. 0%: that is catch rate

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: none (stable)

S120-M-Q3 · question page 2503

AI.EVAL.07 · Choose a measure that matches the goal of a task

How many damaged parcels does that tool catch?

Answer: C. 0

It never flags anything.

✗ A. 100: no
✗ B. 2: it catches none
✗ D. 98: not damaged ones

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S120-M-Q4 · question page 2504

AI.EVAL.07 · Choose a measure that matches the goal of a task

Which measure fits "catch damaged parcels"?

Answer: A. Damaged parcels caught out of all damaged

Catch rate.

✗ B. Number of parcels shipped: not the goal
✗ C. Speed of the tool: not the goal
✗ D. Overall percent correct across every parcel: misleading here

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S120-M-Q5 · question page 2505

AI.EVAL.07 · Choose a measure that matches the goal of a task

Which question checks a measure?

Answer: C. "If this number improves, is the real goal closer?"

Alignment check.

- ✗ A. "Is it easy to compute?": not enough
- ✗ B. "Is the number round?": irrelevant
- ✗ D. "Is this number large enough to look impressive?": not a check

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S120-B1-Q1 · question page 2506

AI.EVAL.07 · Compare measures with judging a basketball player

What does judging only by points miss?

Answer: D. Defense and passing

Other contributions.

- ✗ A. The player's height: not the point
- ✗ B. The jersey number: no
- ✗ C. Nothing: misses a lot

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S120-B1-Q2 · question page 2507

AI.EVAL.07 · Compare measures with judging a basketball player

A player who takes 40 shots to score 20 points shows what?

Answer: A. A number can look better than the reality

Context behind numbers.

- ✗ B. That shots do not matter: they matter
- ✗ C. That defense is bad: not shown

✗ D. That the player is the best on the team: inefficient

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S120-B1-Q3 · question page 2508

AI.EVAL.07 · Compare measures with judging a basketball player

What is the real goal in basketball?

Answer: B. Helping the team win

True objective.

✗ A. Being tallest: no

✗ C. Scoring as many points as possible as an individual, whatever happens to the team: not the real goal

✗ D. Taking the most shots: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S120-B1-Q4 · question page 2509

AI.EVAL.07 · Compare measures with judging a basketball player

Why do coaches use several measures?

Answer: A. One number misses parts of the goal

Fuller picture.

✗ B. To confuse players: no

✗ C. Because one is illegal: no

✗ D. Because coaches enjoy doing extra math: not the reason

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S120-B1-Q5 · question page 2510

AI.EVAL.07 · Compare measures with judging a basketball player

What is the lesson for AI measures?

Answer: D. Connect measures to the real goal

Goal alignment.

- ✗ A. Use whichever single number is the largest, because larger numbers show better results: not the lesson
- ✗ B. Never measure: no
- ✗ C. Measure only speed: too narrow

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S120-F1-Q1 · question page 2511

AI.EVAL.07 · Choose fitting measures in practice

A tool flags fraudulent transactions, which are rare. Which measure is most useful?

Answer: C. Fraud cases caught and how many flags were right

Rare-event measures.

- ✗ A. Number of transactions processed: not the goal
- ✗ B. Average transaction size: not the goal
- ✗ D. Overall percent correct across all transactions: misleading for rare events

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S120-F1-Q2 · question page 2512

AI.EVAL.07 · Choose fitting measures in practice

A writing helper aims to save staff time. Which measure fits best?

Answer: B. Minutes spent editing its drafts

Time saved.

✗ A. Number of fonts used: irrelevant

✗ C. Number of emojis: irrelevant

✗ D. Number of words in each draft: not the goal

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S120-F1-Q3 · question page 2513

AI.EVAL.07 · Choose fitting measures in practice

A tool flags 10 parcels as damaged; 8 really are. What fraction of flags were right?

Answer: A. 8 out of 10

8 / 10 = 80%.

✗ B. 10 out of 8: not a fraction of flags

✗ C. 8 out of 100: wrong total

✗ D. 2 out of 10: reversed

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: none (stable)

S120-F1-Q4 · question page 2514

AI.EVAL.07 · Choose fitting measures in practice

There were 16 damaged parcels in total, and the tool caught 8. What fraction did it catch?

Answer: B. Half

8 / 16 = 1/2.

✗ A. A quarter: arithmetic slip

✗ C. None: it caught 8

✗ D. All of them: missed 8

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: none (stable)

S120-F1-Q5 · question page 2515

AI.EVAL.07 · Choose fitting measures in practice

A translation tool is measured by speed only. What important goal is missing?

Answer: D. Whether translations are accurate

Accuracy matters.

- ✗ A. The tool's color: irrelevant
- ✗ B. The number of languages listed: not the goal
- ✗ C. Nothing is missing: not true

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S120-F2-Q1 · question page 2516

AI.EVAL.07 · Prevent gaming of measures

A help desk is measured on "tickets closed," and staff close tickets without solving problems. What happened?

Answer: C. The number improved without the goal improving

Measure became the target.

- ✗ A. The goal improved a lot, because closing more tickets always means more problems are solved: not here
- ✗ B. Nothing unusual: it is gaming
- ✗ D. The measure is perfect: it misled

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S120-F2-Q2 · question page 2517

AI.EVAL.07 · Prevent gaming of measures

Which measure better reflects real outcomes for a help desk?

Answer: D. Customers who did not need to contact again

Real resolution.

✗ A. Number of tickets opened, since more tickets always mean that the help desk is doing better: not outcomes

✗ B. Average ticket length: not outcomes

✗ C. Number of staff: not outcomes

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S120-F2-Q3 · question page 2518

AI.EVAL.07 · Prevent gaming of measures

Why use more than one measure?

Answer: A. One number can be gamed or miss parts of the goal

Balanced view.

✗ B. To confuse people: no

✗ C. Because using one measure is not allowed and every organization must track exactly ten: not the reason

✗ D. To make reports longer: not the point

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S120-F2-Q4 · question page 2519

AI.EVAL.07 · Prevent gaming of measures

Besides numbers, what should you look at?

Answer: B. Examples behind the numbers

Read real cases.

✗ A. Only the logo: irrelevant

✗ C. Only the price: not enough

✗ D. Nothing else, since numbers always tell the full story about how a tool is performing: not always

Arithmetic checked by code: not applicable. Code example run and its output checked: not